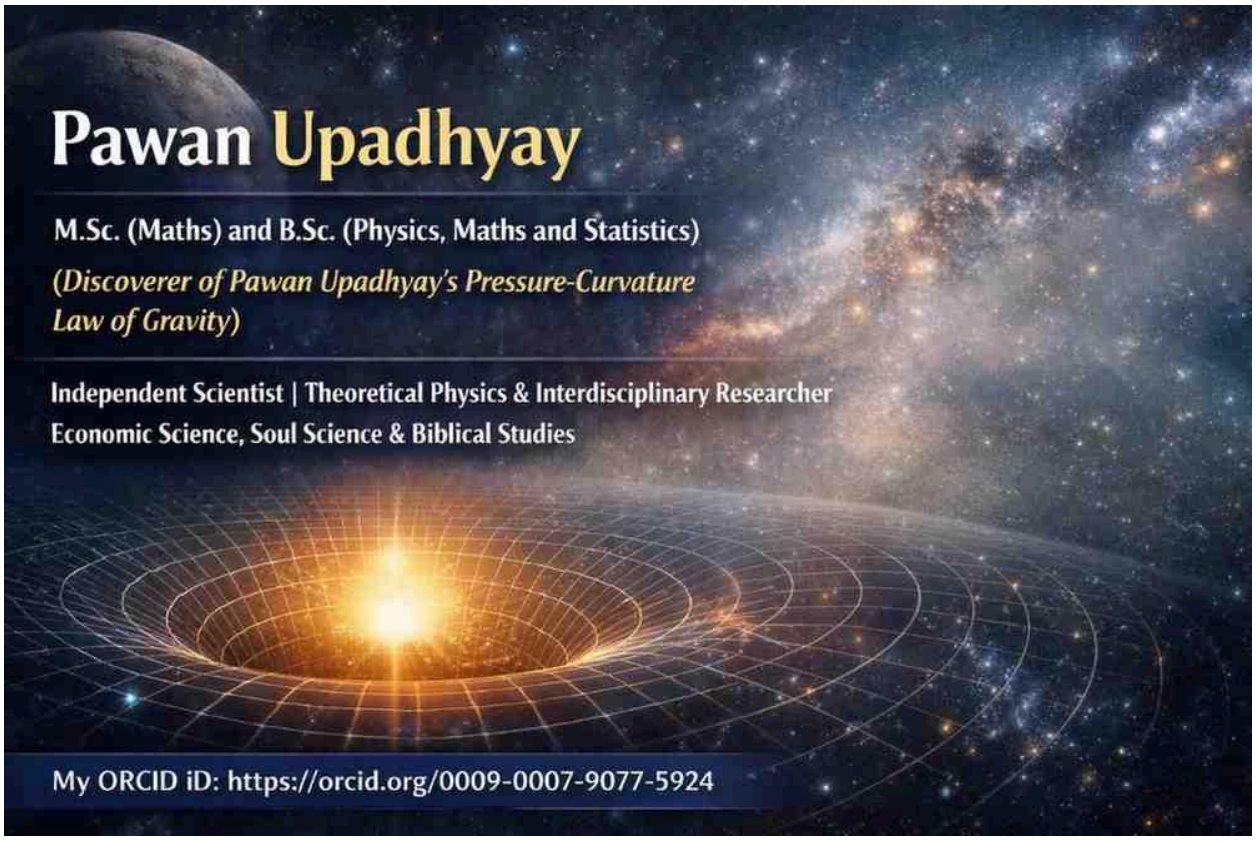
Scalar Gravitational-Pressure Waves on the PPC Metric

A Scalar Extension of the Tensorial PPC Gravitational Field Equation

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Abstract

The PPC formulation of gravity describes the gravitational field through a tensorial equation involving energy density, gravitational pressure, four-velocity, and the spacetime metric. The PPC tensorial formulation is

$$G_{\mu\nu} = (8\pi G/c^4)\,[(E_d + P_g)u_\mu u_\nu + P_g\, g_{\mu\nu}]$$

where E_d is the energy density and P_g is the gravitational pressure. The author's PPC framework also defines the gravitational pressure-energy relation through $P_g = \omega E_d$, with the special case $\omega = 1$ giving $P_g = E_d$.

This paper develops a complementary scalar formulation in which a perturbation of gravitational pressure is represented by a scalar field Φ. The fundamental equilibrium condition is taken to be a spatially uniform gravitational pressure,

$$\nabla P_{g0} = 0.$$

The total gravitational pressure is decomposed as

$$P_g = P_{g0} + \delta P_g$$

and the scalar pressure perturbation is defined by

$$\Phi = \delta P_g.$$

The proposed scalar wave equation on the PPC spacetime metric is then

$$\Box_g \Phi = 0$$

where

$$\Box_g \Phi = 1/\sqrt{|g|}\ \partial_\mu(\sqrt{|g|}\ g^{\mu\nu}\ \partial_\nu\Phi).$$

The resulting framework distinguishes the uniform gravitational-pressure background from its propagating scalar perturbation. The tensorial PPC equation determines the gravitational geometry, while the scalar equation describes the proposed propagation of gravitational-pressure perturbations on that geometry. The paper emphasizes that the scalar wave equation is an additional dynamical hypothesis and is not claimed to follow automatically from the tensorial PPC equation.

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1. Introduction

The concept of pressure plays a central role in the PPC formulation of gravity.

The PPC framework describes gravitational pressure as a physical quantity associated with gravitational energy density. The author's notation defines E_d as energy density and P_g as gravitational pressure, with

$P_g = \omega E_d$.

For $\omega = 1$,

$P_g = E_d$.

The PPC tensorial gravitational equation is represented as

$G_{\mu\nu} = (8\pi G/c^4)\,[(E_d + P_g)u_\mu u_\nu + P_g\, g_{\mu\nu}]$.

This equation is tensorial because its principal quantities carry spacetime indices.

The present paper investigates a different question:

Can a perturbation of gravitational pressure propagate as a scalar wave on the spacetime metric described by the PPC theory?

To investigate this possibility, gravitational pressure is divided into an equilibrium component and a perturbation:

$P_g = P_{g0} + \delta P_g$.

The equilibrium component is assumed to be spatially uniform:

$\nabla P_{g0} = 0$.

The perturbation is then represented by a scalar field:

$\Phi = \delta P_g$.

The proposed scalar propagation equation is

$\Box_g \Phi = 0$.

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2. The PPC Tensorial Gravitational Field

The starting equation is

$G_{\mu\nu} = (8\pi G/c^4)\,[(E_d + P_g)u_\mu u_\nu + P_g\, g_{\mu\nu}]$.

Define the PPC energy-pressure tensor as

$T_{\mu\nu}^{PPC} = [(E_d + P_g)u_\mu u_\nu + P_g\, g_{\mu\nu}]$.

Then

$G_{\mu\nu} = (8\pi G/c^4)T_{\mu\nu}^{PPC}$.

The quantities have the following interpretation:

$G_{\mu\nu}$ = gravitational field tensor

E_d = energy density

P_g = gravitational pressure

u_μ = four-velocity

$g_{\mu\nu}$ = metric tensor.

The metric tensor is therefore an essential component of the tensorial description.

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3. Uniform Gravitational-Pressure Equilibrium

A central condition of the present formulation is

$\nabla P_{g0} = 0$.

This means that the background gravitational pressure is spatially uniform.

Thus,

$P_{g0} = \text{constant}$

with respect to spatial coordinates.

If the PPC gravitational force density is defined as

$F = \nabla P_g$

then the equilibrium background satisfies

$F_0 = \nabla P_{g0} = 0$.

Therefore, a uniform gravitational-pressure background does not produce a pressure-gradient force.

This provides a natural equilibrium state from which perturbations can be studied.

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4. Background Pressure and Perturbation

The total gravitational pressure is decomposed as

$P_g = P_{g0} + \delta P_g$.

Here:

P_{g0} = equilibrium gravitational pressure

δP_g = perturbation of gravitational pressure.

The equilibrium condition is

$\nabla P_{g0} = 0$.

Consequently,

$\nabla P_g = \nabla (P_{g0} + \delta P_g)$.

Therefore,

$\nabla P_g = \nabla P_{g0} + \nabla \delta P_g$.

Since

$\nabla P_{g0} = 0$,

we obtain

$\nabla P_g = \nabla \delta P_g$.

Now define

$\Phi = \delta P_g$.

Then

$\nabla P_g = \nabla \Phi$.

Therefore, the pressure gradient associated with the perturbation is directly represented by the scalar field Φ.

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5. Physical Meaning of Φ

In this paper,

$\Phi \equiv \delta P_g$.

This is an important definition.

The scalar field Φ is not introduced merely as an arbitrary mathematical field. Within the proposed PPC formulation, it represents the **perturbation of gravitational pressure**.

Thus,

$P_g = P_{g0} + \Phi$.

The spatially uniform component is

P_{g0}

while the propagating component is

Φ.

The corresponding pressure gradient is

$\nabla P_g = \nabla \Phi$.

This means that a uniform background can coexist with a non-uniform propagating perturbation.

---

6. The Scalar Wave Equation

A scalar field defined on a curved spacetime is described by the covariant d'Alembertian

$$\Box_g \Phi = 1/\sqrt{|g|}\, \partial_\mu(\sqrt{|g|}\, g^{\mu\nu}\, \partial_\nu \Phi).$$

The proposed source-free scalar gravitational-pressure-wave equation is

$$\Box_g \Phi = 0.$$

Therefore,

$$1/\sqrt{|g|}\, \partial_\mu(\sqrt{|g|}\, g^{\mu\nu}\, \partial_\nu \Phi) = 0.$$

This equation describes the proposed propagation of the scalar gravitational-pressure perturbation on the metric $g_{\mu\nu}$.

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7. Why the Metric Tensor Appears

The scalar field itself has no tensor indices:

$$\Phi = \Phi(x^\mu).$$

However, its propagation takes place on spacetime.

The metric tensor

$$g_{\mu\nu}$$

defines the geometry of that spacetime.

Its inverse

$$g^{\mu\nu}$$

appears directly in the scalar wave operator:

$$\Box_g \Phi = 1/\sqrt{|g|}\, \partial_\mu(\sqrt{|g|}\, g^{\mu\nu}\, \partial_\nu \Phi).$$

Therefore, the proposed formulation can be summarized as

$$g_{\mu\nu} \rightarrow \text{spacetime geometry}$$

$g^{\mu\nu}$ → determines the wave operator

Φ → scalar gravitational-pressure perturbation

$\Box_g \Phi = 0$ → propagation equation.

---

8. The Two PPC Equations

The proposed PPC framework now contains two distinct mathematical equations.

Tensorial gravitational equation

$$G_{\mu\nu} = (8\pi G/c^4)\,[(E_d + P_g)u_\mu u_\nu + P_g\, g_{\mu\nu}]$$

Scalar gravitational-pressure wave equation

$$\Box_g \Phi = 0$$

with

$\Phi = \delta P_g$.

These equations are **not the same equation**.

The first is tensorial.

The second is scalar.

Their connection occurs through the gravitational pressure and the metric.

---

9. Coupling Between the Scalar and Tensorial Sectors

Using

$P_g = P_{g0} + \Phi$

in the PPC tensor equation gives

$G_{\mu\nu} = (8\pi G/c^4)\,[(E_d + P_{g0} + \Phi)u_\mu u_\nu + (P_{g0} + \Phi)g_{\mu\nu}]$.

The perturbation Φ therefore contributes to the PPC energy-pressure tensor.

The perturbation of the PPC tensor is

$\delta T_{\mu\nu}^{PPC} = \Phi(u_\mu u_\nu + g_{\mu\nu})$

under this direct decomposition.

Thus the proposed scalar wave has a route through which it can influence the tensorial gravitational field.

The conceptual chain is

Φ

$\rightarrow \delta P_g$

$\rightarrow \delta T_{\mu\nu}^{PPC}$

$\rightarrow \delta G_{\mu\nu}$

$\rightarrow \delta g_{\mu\nu}$.

This is the proposed PPC scalar-tensor coupling structure.

---

10. Scalar Wave Propagation on the PPC Metric

The scalar wave equation is

$\Box_g \Phi = 0$.

The metric appearing in this equation is the same spacetime metric that occurs in the PPC tensorial gravitational equation.

Thus the two sectors are connected:

PPC tensor equation

↓

$g_{\mu\nu}$

↓

covariant wave operator $\Box_g$

↓

scalar pressure perturbation Φ.

This means that the scalar wave does not propagate independently of spacetime geometry.

Its propagation is determined by the metric.

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11. Flat-Spacetime Limit

When the metric approaches Minkowski spacetime,

$g_{\mu\nu} \to \eta_{\mu\nu}$,

the curved-spacetime operator becomes the ordinary d'Alembertian:

$\Box_g \to \Box$.

Therefore,

$\Box_g \Phi = 0$

reduces to

$\Box\Phi = 0$.

Using the convention employed in the author's scalar-wave work,

$\Box = \nabla^2 - (1/c^2)\partial^2/\partial t^2$.

Therefore,

$\nabla^2\Phi - (1/c^2)\partial^2\Phi/\partial t^2 = 0$.

Rearranging,

$\partial^2\phi/\partial t^2 = c^2\nabla^2\phi$.

The author's existing scalar-wave research establishes this Lorentz-invariant scalar formulation and its Minkowski-spacetime interpretation.

---

12. Uniform Background Does Not Mean No Wave

An important distinction follows from the equilibrium condition.

The background satisfies

$\nabla P_{g0} = 0$.

However, the total pressure is

$P_g = P_{g0} + \phi$.

Therefore,

$\nabla P_g = \nabla\phi$.

Consequently,

$\nabla P_{g0} = 0$

does **not** imply

$\nabla P_g = 0$

when a perturbation exists.

The background is uniform, but the wave perturbation can vary spatially.

This is analogous to the general perturbative structure

$\text{background} + \text{perturbation}$.

Thus the proposed gravitational-pressure wave is associated with Φ, not with the uniform background P_{g0}.

---

13. Plane-Wave Solution

In a locally flat region, consider

$$\Phi = A \exp[i(k \cdot r - \omega t)].$$

Substitution into

$$\Box\Phi = 0$$

gives

$$\omega^2 = c^2 k^2.$$

Therefore,

$$\omega = ck.$$

The phase velocity is

$$v = \omega/k = c.$$

The four-wave-vector satisfies

$$k_\mu k^\mu = 0.$$

The author's scalar-wave work obtains the same plane-wave dispersion relation and null condition for the massless scalar field.

---

14. Pressure Gradient Produced by the Wave

Because

$\Phi = \delta P_g$,

we have

$\nabla P_g = \nabla \Phi$.

For a plane wave,

$\Phi = A \cos(k \cdot r - \omega t)$,

the gradient is

$\nabla \Phi = -A\, k \sin(k \cdot r - \omega t)$.

Therefore,

$F = \nabla P_g = \nabla \Phi$.

Thus the scalar wave produces a spatially varying pressure-gradient field.

The equilibrium state has

$\nabla P_{g0} = 0$,

whereas the wave state has

$\nabla P_g = \nabla \Phi$.

This distinction is fundamental to the proposed model.

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15. Energy-Density Relation

The PPC framework defines

$P_g = \omega E_d$.

For a constant ω, perturbation gives

$\delta P_g = \omega \delta E_d$.

Since

$\Phi = \delta P_g$,

we obtain

$\Phi = \omega \delta E_d$.

Therefore,

$\delta E_d = \Phi / \omega$

for $\omega \neq 0$.

For the special case

$\omega = 1$,

we obtain

$\delta P_g = \delta E_d$.

Consequently,

$\Phi = \delta E_d = \delta P_g$.

The PPC framework therefore permits a particularly simple scalar perturbation interpretation when $\omega = 1$. The underlying pressure-energy relation is part of the author's existing PPC formulation.

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16. Tensorial Einstein Wave Versus Scalar PPC Wave

The distinction between the two wave forms should remain explicit.

Tensorial gravitational wave

$\Box \bar{h}_{\mu\nu} = 0$

Field:

$\bar{h}_{\mu\nu}$

Mathematical type:

Rank-2 tensor

Interpretation:

Metric perturbation

Scalar PPC gravitational-pressure wave

$$\Box_g \Phi = 0$$

Field:

$$\Phi = \delta P_g$$

Mathematical type:

Scalar

Proposed interpretation:

Gravitational-pressure perturbation

The equations have a similar wave-operator structure, but the fields are fundamentally different.

---

17. The Metric Is Not the Scalar

It is also important to distinguish

$g_{\mu\nu}$

from

Φ.

The metric tensor describes spacetime geometry.

The scalar field describes the proposed pressure perturbation.

Therefore,

$g_{\mu\nu} \neq \Phi$.

Instead,

$g_{\mu\nu} \rightarrow$ geometry

and

$\Phi \rightarrow$ scalar pressure perturbation.

The scalar field can nevertheless influence the metric through its appearance in P_g and hence in the PPC tensor equation.

---

18. Perturbative PPC System

For a weak perturbation, write

$$g_{\mu\nu} = g_{\mu\nu 0} + \delta g_{\mu\nu}$$

and

$$P_g = P_{g0} + \Phi.$$

The background satisfies

$$\nabla P_{g0} = 0.$$

The scalar perturbation satisfies

$$\Box_g \Phi = 0.$$

The tensor perturbation is represented by

$$\delta g_{\mu\nu}.$$

Thus the PPC perturbative system contains two possible sectors:

Tensorial sector

$$\delta g_{\mu\nu}$$

Scalar pressure sector

$$\Phi = \delta P_g.$$

The two sectors may interact through the PPC tensor equation.

---

19. Proposed Coupled Equations

The central proposed system is

$$G_{\mu\nu}[g] = (8\pi G/c^4)\,[(E_d + P_{g0} + \Phi)u_\mu u_\nu + (P_{g0} + \Phi)g_{\mu\nu}]$$

and

$$\Box_g \Phi = 0.$$

Together with

$$P_g = P_{g0} + \Phi$$

and

$$\nabla P_{g0} = 0.$$

These equations form the basic mathematical structure of the proposed PPC scalar gravitational-pressure-wave model.

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20. Physical Interpretation of the Coupled System

The proposed interpretation is:

$$P_{g0}$$

is the equilibrium gravitational pressure.

$$\Phi$$

is the perturbation of gravitational pressure.

$$\nabla P_{g0} = 0$$

defines the uniform equilibrium condition.

$\nabla\Phi$

is the pressure gradient associated with the perturbation.

$\Box_g \Phi = 0$

describes the proposed propagation of the perturbation.

$g_{\mu\nu}$

defines the spacetime geometry on which the scalar perturbation propagates.

The PPC tensor equation describes how energy density and gravitational pressure contribute to the gravitational field.

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21. Source-Free and Source-Driven Possibilities

The present paper begins with the source-free equation

$\Box_g \Phi = 0$.

A future generalized version could introduce a scalar source:

$\Box_g \Phi = J_P$

where J_P represents a gravitational-pressure source.

The source would need to be derived from the PPC theory rather than arbitrarily inserted.

Possible source candidates could involve perturbations of

E_d

P_g

u_μ

or other PPC variables.

However, this requires additional theoretical development and is not established by the present formulation.

---

22. Important Theoretical Status

The distinction between derivation and hypothesis is essential.

The following are mathematical identities or definitions within the proposed construction:

$$P_g = P_{g0} + \Phi$$

$$\Phi = \delta P_g$$

$$\nabla P_{g0} = 0.$$

The covariant scalar wave equation

$$\Box_g \Phi = 0$$

is a **proposed dynamical equation** for the gravitational-pressure perturbation.

It does not follow solely from

$$G_{\mu\nu} = (8\pi G/c^4)\,[(E_d + P_g)u_\mu u_\nu + P_g\, g_{\mu\nu}].$$

Therefore, the scientific claim should be stated as:

The PPC framework is extended by hypothesizing that perturbations of gravitational pressure constitute a scalar propagating degree of freedom governed, in the source-free limit, by a covariant scalar wave equation.

This is considerably more precise than claiming that the scalar equation has already been derived from the PPC tensor equation.

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23. Comparison With the Existing Scalar Wave Equation

The author's earlier scalar-wave work establishes the source-free relativistic scalar equation

$$\Box\Phi = 0$$

and its equivalent form

$$\partial^2\Phi/\partial t^2 = c^2\nabla^2\Phi.$$

The present paper gives that mathematical scalar field a new proposed PPC interpretation:

$$\Phi = \delta P_g.$$

Therefore, the relationship is

Existing mathematical scalar wave

$$\Box\Phi = 0$$

↓

PPC physical identification

$$\Phi = \delta P_g$$

↓

PPC gravitational-pressure wave

$$\Box_g \delta P_g = 0.$$

The new step is therefore not the mathematics of the scalar wave operator itself, but its proposed physical identification with a gravitational-pressure perturbation.

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24. Central Mathematical Structure

The complete proposal can be summarized as follows.

Equilibrium

$$P_g = P_{g0}$$

$$\nabla P_{g0} = 0$$

Perturbation

$$P_g = P_{g0} + \Phi$$

Scalar identification

$$\Phi = \delta P_g$$

Pressure gradient

$$\nabla P_g = \nabla \Phi$$

Scalar propagation

$$\Box_g \Phi = 0$$

PPC tensor equation

$$G_{\mu\nu} = (8\pi G/c^4)\,[(E_d + P_g)u_\mu u_\nu + P_g\, g_{\mu\nu}]$$

Coupled form

$$G_{\mu\nu}[g] = (8\pi G/c^4)\,[(E_d + P_{g0} + \Phi)u_\mu u_\nu + (P_{g0} + \Phi)g_{\mu\nu}]$$

This is the central mathematical framework proposed in this paper.

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25. Conceptual Diagram

The theory can be represented as:

$$\text{Uniform background}$$

$$\nabla P_{g0} = 0$$

$$\downarrow$$

$$\text{Gravitational-pressure perturbation}$$

$$\Phi = \delta P_g$$

$$\downarrow$$

$$\text{Pressure gradient}$$

$$\nabla P_g = \nabla \Phi$$

↓

Scalar propagation

$$\Box_g \Phi = 0$$

↓

Perturbation of PPC tensor

$$\delta T_{\mu\nu}^{PPC} = \Phi(u_\mu u_\nu + g_{\mu\nu})$$

↓

Gravitational-field perturbation

$$\delta G_{\mu\nu}$$

↓

Metric response

$$\delta g_{\mu\nu}.$$

This is the proposed scalar-tensor structure of the PPC gravitational-pressure-wave model.

---

26. Conclusions

This paper has developed a proposed scalar extension of the PPC tensorial gravitational framework.

The PPC tensorial equation is

$$G_{\mu\nu} = (8\pi G/c^4)\,[(E_d + P_g)u_\mu u_\nu + P_g\, g_{\mu\nu}].$$

The equilibrium gravitational pressure is assumed to satisfy

$$\nabla P_{g0} = 0.$$

The total pressure is decomposed as

$P_g = P_{g0} + \delta P_g$.

The scalar gravitational-pressure perturbation is defined as

$\Phi = \delta P_g$.

Consequently,

$\nabla P_g = \nabla \Phi$.

The proposed scalar propagation equation on the PPC metric is

$\Box_g \Phi = 0$

with

$\Box_g \Phi = 1/\sqrt{|g|}\, \partial_\mu(\sqrt{|g|}\, g^{\mu\nu} \partial_\nu \Phi)$.

The important result is that **uniform background gravitational pressure does not prevent the existence of a pressure wave**. The background can satisfy

$\nabla P_{g0} = 0$

while the perturbation satisfies

$\nabla \Phi \neq 0$.

Therefore,

$\nabla P_g = \nabla \Phi \neq 0$

during wave propagation.

The scalar and tensorial sectors remain mathematically distinct:

$\Box \bar{h}_{\mu\nu} = 0$

describes the tensorial gravitational-wave sector of linearized GR, whereas

$\Box_g \Phi = 0$

is the proposed scalar gravitational-pressure-wave equation.

The proposed PPC interpretation is therefore:

$g_{\mu\nu}$ → spacetime geometry

P_{g0} → **uniform gravitational-pressure background**

$\Phi = \delta P_g$ → **scalar gravitational-pressure perturbation**

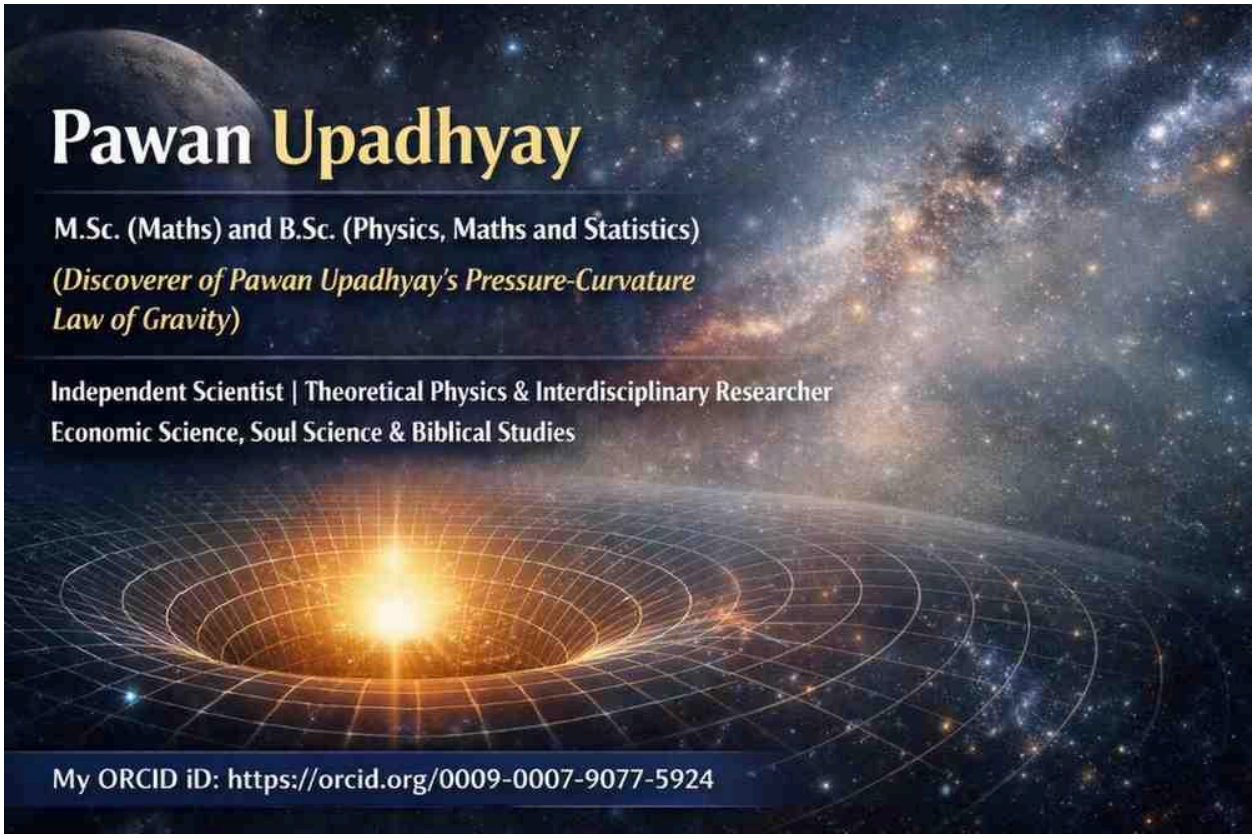
$\Box_g \Phi = 0$ → **propagation of the scalar pressure perturbation**

This provides a clear mathematical foundation for further investigation of gravitational pressure waves within the PPC framework, while keeping the distinction between the established mathematical equations and the new physical hypothesis explicit.

2nd Research Paper

Scalar Gravitational-Pressure Waves in the Weak-Field Limit of the PPC Law of Gravity

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Subject: Scalar gravitational-pressure perturbations and their weak-field propagation

Abstract

Pawan Upadhyay's Pressure Curvature Law (PPC) of Gravity describes gravitation through the combined roles of energy density, gravitational pressure, pressure gradients, and spacetime curvature. In the PPC framework, the energy density is denoted by E_d, the gravitational pressure by P_g, and the relation

$$E_d = \rho c^2$$

is adopted, together with

$$P_g = \omega E_d.$$

The weak-field gravitational potential is governed by the PPC Poisson-type equation

$$\nabla^2 \Phi_g = (4\pi G/c^2)(E_d + 3P_g).$$

This paper develops a separate scalar-wave formulation for perturbations of gravitational pressure. A scalar perturbation field is defined by

$\Phi_P = \delta P_g.$

The proposed dynamical equation is the covariant scalar wave equation

$\Box_g \Phi_P = 0.$

In the weak-field limit, where

$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$

and $|h_{\mu\nu}| << 1$, the equation reduces at leading order to

$\partial^2\Phi_P/\partial t^2 = c^2\nabla^2\Phi_P.$

The paper distinguishes the static scalar gravitational potential Φ_g from the dynamical scalar gravitational-pressure perturbation Φ_P. It further distinguishes this scalar equation from the tensorial gravitational perturbation equation of general relativity. The resulting formulation provides a mathematically clear proposed scalar-wave sector of PPC gravity while retaining the established PPC roles of E_d, P_g, and ∇P_g.

1. Introduction

The PPC Law of Gravity proposes that gravitational phenomena are related to energy density and gravitational pressure. In the PPC formulation, the energy density is written as E_d, while gravitational pressure is written as P_g.

The fundamental PPC relations include

$E_d = \rho c^2$

and

$P_g = \omega E_d.$

The PPC energy-pressure tensor is written as

$T_{\mu\nu}^{PPC} = (E_d + P_g)u_\mu u_\nu + P_g g_{\mu\nu}$

and the corresponding tensorial field equation is

$G_{\mu\nu} = (8\pi G/c^4)T_{\mu\nu}^{PPC}.$

These relations establish two important roles for gravitational pressure. First, P_g contributes directly to the curvature source. Second, its gradient ∇P_g contributes to gravitational-pressure dynamics. The PPC formulation explicitly summarizes this mechanism as

$\rho \rightarrow E_d \rightarrow P_g \rightarrow \nabla P_g \rightarrow T_{\mu\nu} \rightarrow G_{\mu\nu} \rightarrow$ Curvature $\rightarrow$ Geodesic Motion.

The purpose of this paper is to investigate a third, dynamical aspect: whether a perturbation in gravitational pressure can be represented by a scalar wave field.

2. PPC Energy Density and Gravitational Pressure

The PPC framework defines the energy density through

$E_d = \rho c^2$.

It also introduces the gravitational-pressure relation

$P_g = \omega E_d$.

Therefore,

$P_g = \omega \rho c^2$.

The relation between energy density and gravitational pressure is an important component of the PPC model. The supplied PPC material explicitly defines E_d as energy density and P_g as gravitational pressure, with $P_g = \omega E_d$; for $\omega = 1$, $P_g = E_d$.

The pressure gradient is consequently

$\nabla P_g = \nabla(\omega E_d)$.

Applying the product rule,

$\nabla P_g = \omega \nabla E_d + E_d \nabla \omega$.

Thus, variations in gravitational pressure can arise from variations in energy density, the equation-of-state parameter ω, or both.

3. Tensorial PPC Gravitational Field

The PPC energy-momentum tensor is

$T_{\mu\nu}^{PPC} = (E_d + P_g)u_\mu u_\nu + P_g g_{\mu\nu}$.

The PPC field equation is

$G_{\mu v} = (8\pi G/c^4)T_{\mu v}^{PPC}$.

Therefore,

$G_{\mu v} = (8\pi G/c^4)[(E_d + P_g)u_\mu u_v + P_g g_{\mu v}]$.

This is the tensorial gravitational sector of the PPC formulation. The source contains both energy density and gravitational pressure.

The present paper does not replace this tensor equation. Instead, it introduces a scalar dynamical perturbation associated with P_g.

4. Static Weak-Field PPC Equation

In the weak-field limit, the PPC formulation gives the gravitational potential equation

$\nabla^2\Phi_g = (4\pi G/c^2)(E_d + 3P_g)$.

This equation follows from using

$\rho = E_d/c^2$

in the weak-field source containing ρ and gravitational pressure.

The PPC formulation further gives

$P_g = \omega E_d$

and therefore

$\nabla^2\Phi_g = (4\pi G/c^2)E_d(1 + 3\omega)$.

Since

$E_d = \rho c^2$,

this can also be written as

$\nabla^2\Phi_g = 4\pi G\rho(1 + 3\omega)$.

This is the static weak-field gravitational equation.

It is important to distinguish this equation from the scalar wave equation developed below.

5. Static Potential and Scalar Wave Field

Two different scalar quantities should be used to avoid a notational ambiguity.

Let

Φ_g

denote the static gravitational potential.

Let

Φ_P

denote the scalar gravitational-pressure perturbation.

Thus,

$\Phi_g \neq \Phi_P$.

The static field is described by

$\nabla^2 \Phi_g = (4\pi G/c^2)(E_d + 3P_g)$.

The dynamical scalar perturbation is described by

$\Box_g \Phi_P = 0$.

This distinction is essential because the first equation is a Poisson-type static field equation, whereas the second is a hyperbolic propagation equation.

6. Gravitational-Pressure Perturbation

Let the gravitational pressure be decomposed into a background component and a perturbation:

$P_g = P_{g0} + \delta P_g$.

Define the scalar wave field by

$\Phi_P = \delta P_g$.

Therefore,

$P_g = P_{g0} + \Phi_P$.

If the background gravitational pressure is spatially uniform,

$\nabla P_{g0} = 0$.

Consequently,

$\nabla P_g = \nabla \Phi_P$.

This means that the background pressure may be uniform while the perturbation produces a nonzero pressure gradient.

Thus,

$\nabla P_{g0} = 0$

does not imply

$\nabla P_g = 0$

when a perturbation is present.

7. Covariant Scalar Wave Equation

To formulate the scalar perturbation on spacetime, introduce the covariant scalar wave equation

$\Box_g \Phi_P = 0$.

The curved-spacetime scalar wave operator is

$\Box_g \Phi_P = 1/\sqrt{|g|}\, \partial_\mu(\sqrt{|g|}\, g^{\mu\nu}\, \partial_\nu \Phi_P)$.

Here:

$g_{\mu\nu}$ is the spacetime metric,

$g^{\mu\nu}$ is its inverse,

g is the determinant of the metric,

Φ_P is the scalar gravitational-pressure perturbation.

The metric therefore determines the propagation geometry of the scalar field.

8. Weak-Field Metric

For a weak gravitational field, write

$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$

where

$|h_{\mu\nu}| << 1$.

Here:

$\eta_{\mu\nu}$

is the Minkowski background metric and

$h_{\mu\nu}$

is a small metric perturbation.

To leading weak-field order, the scalar wave operator approaches the flat-spacetime d'Alembertian:

$\Box_g \Phi_P \approx \Box \Phi_P$.

Therefore,

$\Box \Phi_P = 0$.

9. Scalar Wave Equation in the Weak-Field Limit

The Minkowski scalar wave equation is

$\Box \Phi_P = 0$.

In three-dimensional spatial notation,

$\nabla^2 \Phi_P - (1/c^2)\partial^2 \Phi_P/\partial t^2 = 0$.

Rearranging,

$\partial^2 \Phi_P/\partial t^2 = c^2 \nabla^2 \Phi_P$

This is the central equation of the present paper.

It describes the proposed propagation of a scalar gravitational-pressure perturbation at wave speed c.

10. Relation Between Pressure and the Scalar Wave

Since

$\Phi_P = \delta P_g$,

we have

$P_g = P_{g0} + \Phi_P$.

Taking the spatial gradient,

$\nabla P_g = \nabla P_{g0} + \nabla \Phi_P$.

For a uniform background,

$\nabla P_{g0} = 0$.

Therefore,

$\nabla P_g = \nabla \Phi_P$.

Thus the scalar wave produces a dynamical gravitational-pressure gradient.

The sequence can be expressed as

$\Phi_P = \delta P_g$

↓

$\nabla \Phi_P = \nabla P_g$

↓

$\nabla P_g \rightarrow$ gravitational-pressure dynamics.

This is consistent with the PPC framework's distinction between gravitational pressure itself and its gradient: P_g contributes to the curvature source, while ∇P_g contributes to pressure-force dynamics.

11. Plane-Wave Solution

Consider a plane scalar wave of the form

$\Phi_P(x,t) = A \exp[i(k\cdot r - \omega t)]$.

Substitution into

$\partial^2\Phi_P/\partial t^2 = c^2\nabla^2\Phi_P$

gives

$\omega^2 = c^2k^2$.

Therefore,

$\omega = ck$.

The phase velocity is

$v = \omega/k = c$.

Hence the proposed scalar gravitational-pressure perturbation is a massless-wave-type disturbance propagating at the speed of light in the weak-field limit.

12. Static and Dynamical Equations Together

The PPC weak-field framework can therefore be represented by two separate equations.

Static gravitational field

$\nabla^2\Phi_g = (4\pi G/c^2)(E_d + 3P_g)$

Scalar gravitational-pressure wave

$\partial^2\Phi_P/\partial t^2 = c^2\nabla^2\Phi_P$

with

$P_g = P_{g0} + \Phi_P$.

The first equation determines the static gravitational potential from the energy-pressure source.

The second determines the propagation of a perturbation in gravitational pressure.

They therefore describe different physical and mathematical aspects of the PPC field.

13. Incorporating the PPC Equation of State

Using

$P_g = \omega E_d$,

the static weak-field equation becomes

$\nabla^2 \Phi_g = (4\pi G/c^2)(1 + 3\omega)E_d$.

If

$E_d = \rho c^2$,

then

$\nabla^2 \Phi_g = 4\pi G\rho(1 + 3\omega)$.

The scalar perturbation can be represented separately as

$\delta P_g = \Phi_P$.

If ω is constant, then a pressure perturbation associated with an energy-density perturbation may be written as

$\delta P_g = \omega \delta E_d$.

Consequently,

$\Phi_P = \omega \delta E_d$.

This provides a possible connection between scalar pressure perturbations and scalar energy-density perturbations.

However, this relation depends on treating ω as the appropriate equation-of-state parameter for the perturbation.

14. Perturbation of the PPC Tensor

The PPC tensor is

$T_{\mu\nu}^{PPC} = (E_d + P_g)u_\mu u_\nu + P_g g_{\mu\nu}$.

Introduce

$P_g = P_{g0} + \Phi_P$.

Then

$T_{\mu\nu}^{PPC} = [E_d + P_{g0} + \Phi_P]u_\mu u_\nu + [P_{g0} + \Phi_P]g_{\mu\nu}$.

The part directly proportional to the scalar pressure perturbation is

$\delta T_{\mu\nu}^{PPC} = \Phi_P(u_\mu u_\nu + g_{\mu\nu})$

when the other quantities are held fixed at the perturbative level.

Thus the scalar pressure field can enter the tensorial PPC source through

$\Phi_P \rightarrow \delta T_{\mu\nu}^{PPC} \rightarrow \delta G_{\mu\nu}$.

This provides a possible coupling between the scalar pressure-wave sector and the tensorial gravitational geometry.

15. Tensorial and Scalar Forms

The distinction between tensorial and scalar wave equations is fundamental.

Tensorial gravitational perturbation

In a linearized metric description, the gravitational perturbation is represented by a tensor

$h_{\mu\nu}$

or the trace-reversed tensor

$\bar{h}_{\mu\nu}$.

The vacuum tensorial wave equation takes the form

$\Box \bar{h}_{\mu\nu} = 0$.

Scalar gravitational-pressure perturbation

The PPC scalar pressure perturbation is

$\Phi_P = \delta P_g$

and its proposed wave equation is

$\Box \Phi_P = 0$.

Thus,

$\Box \bar{h}_{\mu\nu} = 0$

and

□Φ_P = 0

have similar mathematical wave-operator structure but describe different fields.

The first is tensorial.

The second is scalar.

They must not be identified with one another.

16. Metric Tensor and Scalar Wave

The scalar wave is not itself a metric tensor.

The relationship is instead

g_μv → □_g → Φ_P.

The metric determines the covariant propagation operator acting on the scalar field.

In the weak-field approximation,

g_μv = η_μv + h_μv

and therefore

□_g Φ_P ≈ □Φ_P.

Consequently,

□Φ_P = 0

is obtained at leading weak-field order.

This gives a clear mathematical interpretation:

the scalar gravitational-pressure wave propagates on the weakly curved spacetime described by the PPC metric.

17. Coupled PPC Scalar-Tensor Structure

The proposed combined structure can be summarized as

E_d, P_g → T_μv^PPC → G_μv → g_μv

and independently,

$\delta P_g = \Phi_P \rightarrow \Box_g \Phi_P = 0.$

The two sectors can then be coupled:

$\Phi_P \rightarrow \delta P_g \rightarrow \delta T_{\mu\nu}^{PPC} \rightarrow \delta G_{\mu\nu} \rightarrow \delta g_{\mu\nu}.$

Thus the proposed PPC structure contains:

Tensorial sector

$G_{\mu\nu} = (8\pi G/c^4)[(E_d + P_g)u_\mu u_\nu + P_g g_{\mu\nu}]$

Scalar sector

$\Box_g \Phi_P = 0$

Weak-field scalar sector

$\partial^2 \Phi_P/\partial t^2 = c^2 \nabla^2 \Phi_P.$

This is a scalar-tensor-like mathematical structure within the proposed PPC framework.

18. Static Limit of the Scalar Sector

The scalar wave equation is

$\partial^2 \Phi_P/\partial t^2 = c^2 \nabla^2 \Phi_P.$

For a strictly static perturbation,

$\partial \Phi_P/\partial t = 0.$

Therefore,

$\partial^2 \Phi_P/\partial t^2 = 0.$

The wave equation becomes

$\nabla^2 \Phi_P = 0.$

Thus, the homogeneous static limit of the scalar wave equation is a Laplace equation.

This is distinct from the sourced PPC gravitational-potential equation

$\nabla^2 \Phi_g = (4\pi G/c^2)(E_d + 3P_g)$.

The latter contains the gravitational source; the former describes the homogeneous static limit of the freely propagating scalar perturbation.

19. Role of the Pressure Gradient

The PPC framework assigns a dynamical role to the pressure gradient.

The pressure force is represented as

$F = -\nabla P_g$.

With

$P_g = P_{g0} + \Phi_P$

and

$\nabla P_{g0} = 0$,

we obtain

$F = -\nabla \Phi_P$.

Therefore, the scalar pressure perturbation generates a pressure-gradient force.

The PPC framework also gives the relation

$a^\mu = -[(g^{\mu\nu} + u^\mu u^\nu)\nabla_\nu P_g]/(E_d + P_g)$

for the pressure-gradient contribution to four-acceleration.

Consequently, the scalar perturbation can be connected to dynamical motion through

$\Phi_P \rightarrow \nabla \Phi_P \rightarrow \nabla P_g \rightarrow a^\mu$.

20. Physical Interpretation

The proposed scalar gravitational-pressure wave can be interpreted as a propagating perturbation of the PPC gravitational-pressure field.

The physical sequence is

$E_d \rightarrow P_g \rightarrow \delta P_g \rightarrow \Phi_P \rightarrow \nabla \Phi_P \rightarrow$ pressure-gradient dynamics.

The static gravitational field is instead represented by

$E_d + 3P_g \rightarrow \Phi_g$.

Therefore, the PPC framework contains two conceptually different scalar quantities:

Φ_g — static gravitational potential

Φ_P — dynamical gravitational-pressure perturbation.

The distinction prevents the static Poisson equation from being incorrectly identified with the dynamical wave equation.

21. Weak-Field Approximation

The weak-field assumption is

$|h_{\mu\nu}| << 1$.

The metric is therefore

$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$.

The scalar field equation is initially

$\Box_g \Phi_P = 0$.

Expanding around Minkowski spacetime gives, schematically,

$(\Box + \delta\Box)\Phi_P = 0$

where $\delta\Box$ represents corrections produced by the weak gravitational geometry.

At leading order,

$\delta\Box\Phi_P$

is neglected, yielding

$\Box\Phi_P = 0$.

Therefore,

$\partial^2\Phi_P/\partial t^2 = c^2\nabla^2\Phi_P$.

Higher-order corrections would require solving the scalar equation together with the perturbed PPC metric rather than using the flat-space approximation.

22. Proposed Fundamental Equations

The proposed scalar-wave extension of PPC gravity can therefore be summarized by the following equations.

PPC energy density:-

$E_d = \rho c^2$

PPC gravitational pressure:-

$P_g = \omega E_d$

PPC tensor :-

$T_{\mu\nu}^{PPC} = (E_d + P_g)u_\mu u_\nu + P_g g_{\mu\nu}$

PPC gravitational field equation:-

$G_{\mu\nu} = (8\pi G/c^4)T_{\mu\nu}^{PPC}$

Static weak-field gravitational equation:-

$\nabla^2\Phi_g = (4\pi G/c^2)(E_d + 3P_g)$

Pressure perturbation:-

$P_g = P_{g0} + \Phi_P$

Scalar pressure-wave equation on the metric:-

$\Box_g\Phi_P = 0$

Weak-field scalar equation:-

$\Box\Phi_P = 0$

Explicit weak-field wave equation:-

$\partial^2\Phi_P/\partial t^2 = c^2\nabla^2\Phi_P$

Pressure-gradient relation:-

$\nabla P_g = \nabla \Phi_P$

for

$\nabla P_{g0} = 0$.

23. Central Mathematical Structure

The complete proposed weak-field PPC structure can be written as:

$\rho \rightarrow E_d = \rho c^2$

$E_d \rightarrow P_g = \omega E_d$

$E_d + P_g \rightarrow T_{\mu\nu}^{PPC}$

$T_{\mu\nu}^{PPC} \rightarrow G_{\mu\nu} \rightarrow g_{\mu\nu}$

while the perturbative scalar sector is

$P_g = P_{g0} + \Phi_P$

$\Phi_P = \delta P_g$

$\Box_g \Phi_P = 0$

and in the weak-field limit,

$\partial^2 \Phi_P / \partial t^2 = c^2 \nabla^2 \Phi_P$.

The static field remains

$\nabla^2 \Phi_g = (4\pi G/c^2)(E_d + 3P_g)$.

This provides a separation between static curvature generation and dynamic scalar pressure propagation.

24. Theoretical Status of the Scalar Wave Equation

A crucial distinction must be made.

The PPC tensor equation by itself establishes the energy-pressure source:

$G_{\mu\nu} = (8\pi G/c^4)[(E_d + P_g)u_\mu u_\nu + P_g g_{\mu\nu}]$.

The scalar equation

$\Box_g \Phi_P = 0$

is an additional dynamical postulate for the perturbation

$\Phi_P = \delta P_g$.

It does not follow automatically from the static weak-field equation

$\nabla^2 \Phi_g = (4\pi G/c^2)(E_d + 3P_g)$.

Therefore, the scientifically precise statement is:

The PPC framework can be extended by postulating that gravitational-pressure perturbations constitute a massless scalar field obeying the covariant wave equation $\Box_g \Phi_P = 0$. In the weak-field limit this reduces to $\partial^2 \Phi_P / \partial t^2 = c^2 \nabla^2 \Phi_P$.

This makes clear which part belongs to the existing PPC formulation and which part is the proposed scalar-wave extension.

25. Conclusions

This paper develops a separate scalar-wave formulation for the PPC Law of Gravity.

The PPC energy density is

$E_d = \rho c^2$

and gravitational pressure is

$P_g = \omega E_d$.

The static weak-field gravitational equation is

$\nabla^2 \Phi_g = (4\pi G/c^2)(E_d + 3P_g)$.

A gravitational-pressure perturbation is defined as

$\Phi_P = \delta P_g$.

The corresponding proposed covariant scalar equation is

$\Box_g \Phi_P = 0$.

Under the weak-field approximation,

$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$

with

$|h_{\mu\nu}| \ll 1$,

the scalar equation reduces to

$\Box \Phi_P = 0$.

In ordinary three-dimensional notation this becomes

$\partial^2 \Phi_P / \partial t^2 = c^2 \nabla^2 \Phi_P$.

The resulting formulation distinguishes three mathematically different objects:

Φ_g — static gravitational potential,

$h_{\mu\nu}$ — tensorial metric perturbation,

Φ_P — scalar gravitational-pressure perturbation.

The PPC pressure itself contributes to the gravitational curvature source, while its gradient contributes to pressure-force dynamics. This distinction is already present in the PPC formulation.

The proposed scalar-wave extension adds a dynamical propagation law for perturbations of P_g. It therefore provides a possible scalar sector alongside the tensorial PPC gravitational field.

Final Proposed PPC Scalar-Wave Equation

For the paper, the central equation can be stated simply as:

Gravitational-pressure perturbation

$\Phi_P = \delta P_g$

Covariant form

$\Box_g \Phi_P = 0$

Weak-field form

$\Box \Phi_P = 0$

Explicit wave equation

$\partial^2 \Phi_P / \partial t^2 = c^2 \nabla^2 \Phi_P$

with

$P_g = P_{g0} + \Phi_P$

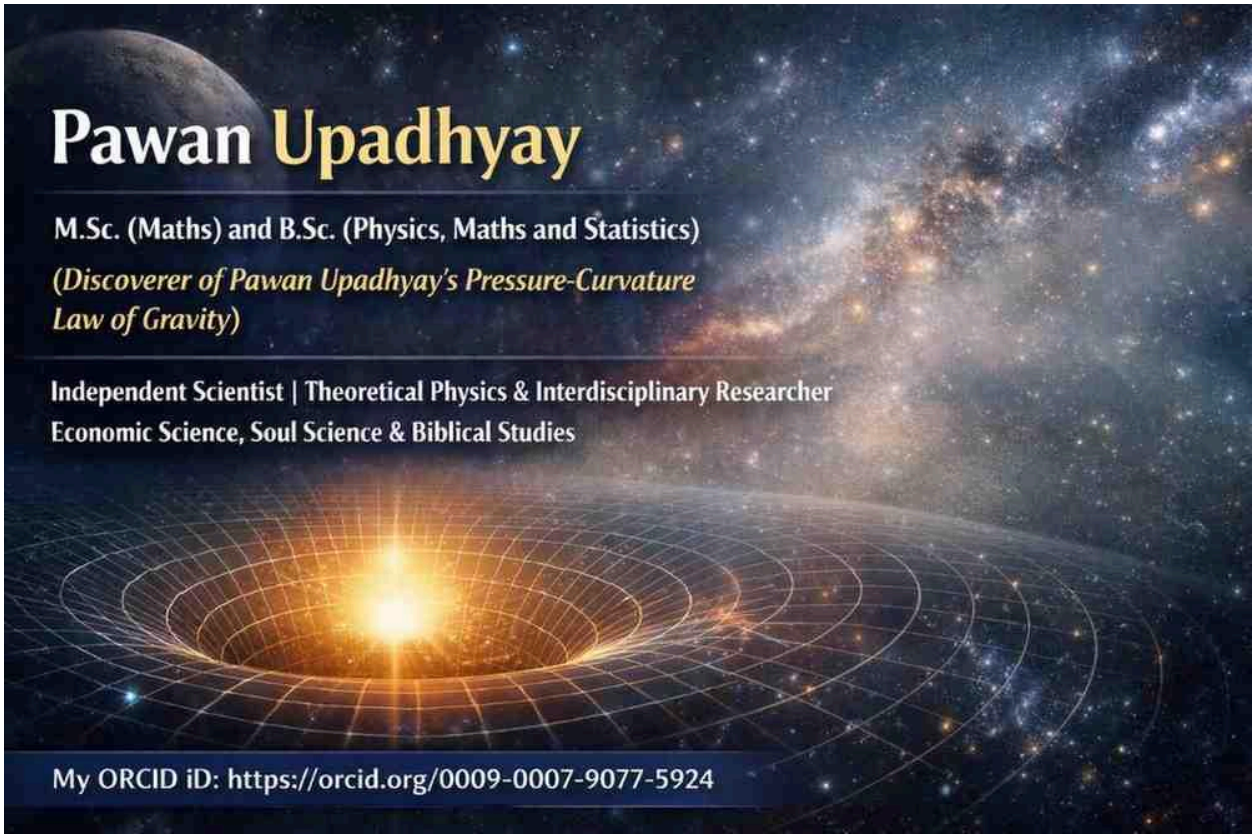
and, for a uniform background,

$\nabla P_g = \nabla \Phi_P$.

This is the cleanest separate formulation of the scalar gravitational-pressure wave in the weak-field limit of the PPC Law of Gravity.

SCALAR GRAVITATIONAL-PRESSURE WAVES IN THE WEAK-FIELD LIMIT OF THE PPC LAW OF GRAVITY

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ABSTRACT

The Pressure Curvature Law (PPC) of Gravity describes gravitation through the combined roles of energy density, gravitational pressure, pressure gradients, and spacetime curvature. In the PPC framework, the energy density is denoted by E_d, the gravitational pressure by P_g, and the relation

$$E_d = \rho c^2$$

is adopted, together with

$$P_g = \omega E_d.$$

The weak-field gravitational potential is governed by the PPC equation

$$\nabla^2 \Phi_g = (4\pi G/c^2)(E_d + 3P_g).$$

This paper develops a separate scalar-wave formulation for perturbations of gravitational pressure. A scalar perturbation field is defined by

$$\Phi_P = \delta P_g.$$

The proposed dynamical equation is

$\Box_g \Phi_P = 0.$

In the weak-field limit, where

$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$

and

$|h_{\mu\nu}| << 1,$

the scalar wave equation reduces, to leading order, to

$\Box \Phi_P = 0.$

Therefore,

$\partial^2 \Phi_P / \partial t^2 = c^2 \nabla^2 \Phi_P.$

The paper distinguishes the static gravitational potential Φ_g from the dynamical scalar gravitational-pressure perturbation Φ_P. It also distinguishes the scalar wave equation from the tensorial gravitational perturbation equation. The proposed formulation provides a mathematical scalar-wave sector for PPC gravity while retaining the PPC roles of E_d, P_g, and ∇P_g.

1. INTRODUCTION

The PPC Law of Gravity proposes that gravitational phenomena are related to energy density and gravitational pressure. In the PPC formulation, the energy density is denoted by E_d and gravitational pressure is denoted by P_g.

The fundamental PPC relations are

$E_d = \rho c^2$

and

$P_g = \omega E_d.$

The PPC energy-pressure tensor is

$T_{\mu\nu}^{PPC} = (E_d + P_g) u_\mu u_\nu + P_g g_{\mu\nu}$

and the tensorial field equation is

$G_{\mu\nu} = (8\pi G / c^4) T_{\mu\nu}^{PPC}.$

Within the PPC framework, gravitational pressure has two distinct roles. First, P_g contributes to the curvature source. Second, its gradient ∇P_g contributes to pressure-force dynamics.

The PPC mechanism can therefore be summarized as

$\rho \rightarrow E_d \rightarrow P_g \rightarrow \nabla P_g \rightarrow T_{\mu\nu} \rightarrow G_{\mu\nu} \rightarrow$ Curvature $\rightarrow$ Geodesic Motion.

The purpose of this paper is to develop a separate dynamical description of perturbations in gravitational pressure.

2. PPC ENERGY DENSITY AND GRAVITATIONAL PRESSURE

The PPC energy density is

$E_d = \rho c^2$.

The gravitational pressure is

$P_g = \omega E_d$.

Therefore,

$P_g = \omega \rho c^2$.

If $\omega = 1$, then

$P_g = E_d$.

The pressure gradient is

$\nabla P_g = \nabla(\omega E_d)$.

Applying the product rule,

$\nabla P_g = \omega \nabla E_d + E_d \nabla \omega$.

Therefore, spatial variations in gravitational pressure can arise from variations in energy density, variations in ω, or both.

3. PPC TENSORIAL GRAVITATIONAL FIELD

The PPC energy-pressure tensor is

$T_{\mu\nu}^{PPC} = (E_d + P_g)u_\mu u_\nu + P_g g_{\mu\nu}$.

The corresponding field equation is

$G_{\mu\nu} = (8\pi G/c^4)[(E_d + P_g)u_\mu u_\nu + P_g g_{\mu\nu}]$.

This equation represents the tensorial gravitational sector of the PPC formulation.

The present paper does not replace the tensorial PPC equation. Instead, it introduces a scalar dynamical perturbation associated with gravitational pressure.

4. STATIC WEAK-FIELD PPC EQUATION

The PPC weak-field gravitational equation is

$$\nabla^2\Phi_g = (4\pi G/c^2)(E_d + 3P_g).$$

Using

$$P_g = \omega E_d,$$

we obtain

$$\nabla^2\Phi_g = (4\pi G/c^2)E_d(1 + 3\omega).$$

Since

$$E_d = \rho c^2,$$

the equation becomes

$$\nabla^2\Phi_g = 4\pi G\rho(1 + 3\omega).$$

This is the static weak-field gravitational equation of the PPC formulation.

5. STATIC GRAVITATIONAL POTENTIAL AND SCALAR WAVE FIELD

Two different scalar quantities are used in this paper.

Φ_g = static gravitational potential

Φ_P = scalar gravitational-pressure perturbation.

Therefore,

$$\Phi_g \neq \Phi_P.$$

The static gravitational field is described by

$$\nabla^2\Phi_g = (4\pi G/c^2)(E_d + 3P_g).$$

The dynamical scalar pressure perturbation is described by

$$\Box_g\Phi_P = 0.$$

The two equations have different mathematical and physical meanings.

6. GRAVITATIONAL-PRESSURE PERTURBATION

Let the gravitational pressure be decomposed into a background pressure and a perturbation:

$P_g = P_{g0} + \delta P_g$.

Define the scalar pressure-wave field as

$\Phi_P = \delta P_g$.

Therefore,

$P_g = P_{g0} + \Phi_P$.

If the background gravitational pressure is spatially uniform, then

$\nabla P_{g0} = 0$.

Consequently,

$\nabla P_g = \nabla \Phi_P$.

Thus, a uniform background pressure does not prevent the existence of a pressure wave. The perturbation can still have

$\nabla \Phi_P \neq 0$.

7. COVARIANT SCALAR WAVE EQUATION

The proposed scalar gravitational-pressure wave equation on the PPC metric is

$\Box_g \Phi_P = 0$.

The curved-spacetime scalar wave operator is

$\Box_g \Phi_P = 1/\sqrt{|g|}\, \partial_\mu(\sqrt{|g|}\, g^{\mu\nu}\, \partial_\nu \Phi_P)$.

Here,

$g_{\mu\nu}$ = spacetime metric

$g^{\mu\nu}$ = inverse metric

g = determinant of the metric

Φ_P = scalar gravitational-pressure perturbation.

The metric therefore determines the geometrical propagation of the scalar field.

8. WEAK-FIELD METRIC

In the weak-field limit, the metric is written as

$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$

where

$|h_{\mu\nu}| << 1$.

Here $\eta_{\mu\nu}$ is the Minkowski metric and $h_{\mu\nu}$ represents a small metric perturbation.

To leading weak-field order, the covariant scalar wave operator approaches the flat-spacetime wave operator:

$\Box_g \Phi_P \approx \Box \Phi_P$.

Therefore,

$\Box \Phi_P = 0$.

9. SCALAR WAVE EQUATION IN THE WEAK-FIELD LIMIT

The Minkowski scalar wave equation is

$\Box \Phi_P = 0$.

In three-dimensional spatial notation,

$\nabla^2 \Phi_P - (1/c^2)\partial^2 \Phi_P/\partial t^2 = 0$.

Rearranging,

$\partial^2 \Phi_P/\partial t^2 = c^2 \nabla^2 \Phi_P$.

Therefore, the central scalar gravitational-pressure wave equation is

$\partial^2 \Phi_P/\partial t^2 = c^2 \nabla^2 \Phi_P$.

This equation describes the proposed propagation of a scalar gravitational-pressure perturbation at wave speed c in the weak-field limit.

10. PLANE-WAVE SOLUTION

Consider a plane-wave solution

$\Phi_P = A \exp[i(k \cdot r - \omega t)]$.

Substitution into

$\partial^2 \Phi_P / \partial t^2 = c^2 \nabla^2 \Phi_P$

gives

$\omega^2 = c^2 k^2$.

Therefore,

$\omega = ck$.

The phase velocity is

$v = \omega / k = c$.

Thus, under the proposed scalar-wave hypothesis, the gravitational-pressure perturbation propagates at the speed of light in the weak-field limit.

11. RELATION BETWEEN P_g AND THE SCALAR WAVE

Since

$\Phi_P = \delta P_g$,

the total gravitational pressure is

$P_g = P_{g0} + \Phi_P$.

Taking the spatial gradient gives

$\nabla P_g = \nabla P_{g0} + \nabla \Phi_P$.

For a uniform background,

$\nabla P_{g0} = 0$.

Therefore,

$\nabla P_g = \nabla \Phi_P$.

The scalar wave consequently produces a dynamic gravitational-pressure gradient.

The sequence is

$\Phi_P = \delta P_g$

↓

$\nabla \Phi_P = \nabla P_g$

↓

$\nabla P_g \rightarrow$ gravitational-pressure dynamics.

12. PPC EQUATION OF STATE

The PPC equation of state is

$P_g = \omega E_d$.

For a perturbation,

$\delta P_g = \delta\omega\, E_d + \omega \delta E_d$.

If ω is constant, then

$\delta\omega = 0$

and therefore

$\delta P_g = \omega \delta E_d$.

Since

$\Phi_P = \delta P_g$,

we obtain

$\Phi_P = \omega \delta E_d$.

Thus, for constant ω, the scalar gravitational-pressure perturbation can be directly related to an energy-density perturbation.

13. PERTURBATION OF THE PPC TENSOR

The PPC tensor is

$T_{\mu\nu}^{PPC} = (E_d + P_g)u_\mu u_\nu + P_g g_{\mu\nu}$.

Using

$P_g = P_{g0} + \Phi_P$,

we obtain

$T_{\mu\nu}^{PPC} = [E_d + P_{g0} + \Phi_P]u_\mu u_\nu + [P_{g0} + \Phi_P]g_{\mu\nu}$.

If E_d, u_μ, and $g_{\mu\nu}$ are held fixed at the perturbative level, the direct scalar pressure contribution is

$\delta T_{\mu\nu}^{PPC} = \Phi_P(u_\mu u_\nu + g_{\mu\nu})$.

Therefore, the scalar perturbation can enter the tensorial PPC source through

$\Phi_P \rightarrow \delta T_{\mu\nu}^{PPC} \rightarrow \delta G_{\mu\nu}$.

This provides a possible coupling between the scalar pressure-wave sector and the tensorial gravitational geometry.

14. TENSORIAL AND SCALAR WAVE EQUATIONS

The tensorial and scalar wave equations describe different fields.

Tensorial gravitational perturbation:

$\Box \bar{h}_{\mu\nu} = 0$.

Here $\bar{h}_{\mu\nu}$ is a tensorial metric perturbation.

Scalar gravitational-pressure perturbation:

$\Box \Phi_P = 0$.

Here Φ_P is a scalar field.

Therefore,

$\Box \bar{h}_{\mu\nu} = 0$

and

$\Box \Phi_P = 0$

have the same mathematical wave-operator structure but represent different physical degrees of freedom.

The tensorial field should not be identified with the scalar pressure field.

15. SCALAR WAVE ON THE METRIC

The scalar pressure field is not itself a metric tensor.

The relationship is

$g_{\mu\nu} \rightarrow \Box_g \rightarrow \Phi_P$.

The metric determines the covariant wave operator acting on the scalar field.

In the weak-field approximation,

$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$

and

$\Box_g \Phi_P \approx \Box \Phi_P$.

Therefore,

$\Box \Phi_P = 0$

and

$\partial^2 \Phi_P / \partial t^2 = c^2 \nabla^2 \Phi_P$.

Thus, the scalar gravitational-pressure wave propagates on the weakly curved spacetime described by the PPC metric.

16. STATIC LIMIT OF THE SCALAR WAVE

The scalar wave equation is

$\partial^2 \Phi_P / \partial t^2 = c^2 \nabla^2 \Phi_P$.

For a strictly static perturbation,

$\partial \Phi_P / \partial t = 0$.

Therefore,

$\partial^2 \Phi_P / \partial t^2 = 0$.

The equation reduces to

$\nabla^2 \Phi_P = 0$.

This is the homogeneous static limit of the scalar wave equation.

It should not be confused with the sourced PPC weak-field gravitational equation

$\nabla^2 \Phi_g = (4\pi G/c^2)(E_d + 3P_g)$.

17. PRESSURE-GRADIENT FORCE

The PPC pressure-force relation is

$F = -\nabla P_g$.

Using

$P_g = P_{g0} + \Phi_P$

and

$\nabla P_{g0} = 0$,

we obtain

$F = -\nabla \Phi_P$.

Thus, the scalar pressure perturbation produces a pressure-gradient force.

The corresponding dynamical sequence is

$\Phi_P \rightarrow \nabla \Phi_P \rightarrow \nabla P_g \rightarrow F$.

In the PPC framework, the pressure gradient also contributes to four-acceleration according to

$a^\mu = -[(g^{\mu v} + u^\mu u^v)\nabla_v P_g]/(E_d + P_g)$.

Therefore, the scalar perturbation may be connected to matter dynamics through

$\Phi_P \rightarrow \nabla \Phi_P \rightarrow \nabla P_g \rightarrow a^\mu$.

18. STATIC AND DYNAMIC PPC EQUATIONS

The PPC weak-field formulation can be represented by two distinct equations.

STATIC GRAVITATIONAL FIELD:

$\nabla^2 \Phi_g = (4\pi G/c^2)(E_d + 3P_g)$.

SCALAR GRAVITATIONAL-PRESSURE WAVE:

$\partial^2 \Phi_P / \partial t^2 = c^2 \nabla^2 \Phi_P$.

The first equation describes the static gravitational potential.

The second equation describes the propagation of a perturbation in gravitational pressure.

Therefore, the static and dynamic equations should not be treated as identical equations.

19. COUPLED SCALAR-TENSOR PPC STRUCTURE

The complete proposed structure can be written as

$E_d, P_g \rightarrow T_{\mu\nu}^{PPC} \rightarrow G_{\mu\nu} \rightarrow g_{\mu\nu}$

and

$\delta P_g = \Phi_P \rightarrow \Box_g \Phi_P = 0.$

The scalar perturbation can also modify the PPC tensor:

$\Phi_P \rightarrow \delta T_{\mu\nu}^{PPC} \rightarrow \delta G_{\mu\nu} \rightarrow \delta g_{\mu\nu}.$

Thus, the proposed PPC framework contains a tensorial gravitational sector and a scalar pressure-wave sector.

TENSORIAL SECTOR:

$G_{\mu\nu} = (8\pi G/c^4)[(E_d + P_g)u_\mu u_\nu + P_g g_{\mu\nu}]$

SCALAR SECTOR:

$\Box_g \Phi_P = 0$

WEAK-FIELD SCALAR SECTOR:

$\partial^2 \Phi_P / \partial t^2 = c^2 \nabla^2 \Phi_P.$

20. PHYSICAL INTERPRETATION

The scalar gravitational-pressure wave represents a propagating perturbation of the gravitational-pressure field.

The proposed physical sequence is

$E_d \rightarrow P_g \rightarrow \delta P_g \rightarrow \Phi_P \rightarrow \nabla \Phi_P \rightarrow \nabla P_g \rightarrow$ gravitational-pressure dynamics.

The static gravitational field is represented by

$E_d + 3P_g \rightarrow \Phi_g.$

Therefore, two different scalar quantities are used:

Φ_g = static gravitational potential

Φ_P = scalar gravitational-pressure perturbation.

This distinction is important because the static gravitational potential and the dynamic pressure perturbation have different mathematical equations.

21. WEAK-FIELD APPROXIMATION

The weak-field condition is

$|h_{\mu\nu}| \ll 1$.

The metric is

$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$.

The scalar equation is initially

$\Box_g \Phi_P = 0$.

Expanding around Minkowski spacetime gives

$(\Box + \delta\Box)\Phi_P = 0$.

Here $\delta\Box$ represents corrections caused by the weak gravitational geometry.

At leading order, the correction is neglected:

$\delta\Box\Phi_P \approx 0$.

Therefore,

$\Box\Phi_P = 0$.

Consequently,

$\partial^2\Phi_P/\partial t^2 = c^2\nabla^2\Phi_P$.

Higher-order corrections would require solving the scalar equation together with the perturbed PPC metric.

22. COMPLETE SET OF PROPOSED EQUATIONS

The proposed PPC scalar-wave formulation can be summarized as follows.

PPC ENERGY DENSITY:

$E_d = \rho c^2$

PPC GRAVITATIONAL PRESSURE:

$P_g = \omega E_d$

PPC ENERGY-PRESSURE TENSOR:

$T_{\mu\nu}^{PPC} = (E_d + P_g)u_\mu u_\nu + P_g g_{\mu\nu}$

PPC FIELD EQUATION:

$G_{\mu\nu} = (8\pi G/c^4)T_{\mu\nu}^{PPC}$

STATIC WEAK-FIELD EQUATION:

$\nabla^2\Phi_g = (4\pi G/c^2)(E_d + 3P_g)$

EQUATION OF STATE SUBSTITUTION:

$\nabla^2\Phi_g = (4\pi G/c^2)E_d(1 + 3\omega)$

SCALAR PRESSURE PERTURBATION:

$\Phi_P = \delta P_g$

TOTAL PRESSURE:

$P_g = P_{g0} + \Phi_P$

COVARIANT SCALAR WAVE EQUATION:

$\Box_g\Phi_P = 0$

WEAK-FIELD SCALAR EQUATION:

$\Box\Phi_P = 0$

EXPLICIT WEAK-FIELD WAVE EQUATION:

$\partial^2\Phi_P/\partial t^2 = c^2\nabla^2\Phi_P$

PRESSURE GRADIENT:

$\nabla P_g = \nabla\Phi_P$

for

$\nabla P_{g0} = 0.$

23. CENTRAL MATHEMATICAL STRUCTURE

The central PPC structure can be written as

$\rho \rightarrow E_d = \rho c^2$

$E_d \rightarrow P_g = \omega E_d$

$E_d + P_g \rightarrow T_{\mu\nu}^{PPC}$

$T_{\mu\nu}^{PPC} \rightarrow G_{\mu\nu} \rightarrow g_{\mu\nu}$

The scalar perturbation sector is

$P_g = P_{g0} + \Phi_P$

$\Phi_P = \delta P_g$

$\Box_g \Phi_P = 0$

and in the weak-field limit,

$\partial^2 \Phi_P / \partial t^2 = c^2 \nabla^2 \Phi_P$.

The static gravitational field remains

$\nabla^2 \Phi_g = (4\pi G/c^2)(E_d + 3P_g)$.

Therefore, the proposed PPC framework separates static gravitational curvature from dynamic scalar gravitational-pressure propagation.

24. THEORETICAL STATUS

The PPC tensor equation provides the energy-pressure source of the gravitational field:

$G_{\mu\nu} = (8\pi G/c^4)[(E_d + P_g)u_\mu u_\nu + P_g g_{\mu\nu}]$.

The scalar equation

$\Box_g \Phi_P = 0$

is an additional dynamical hypothesis for the perturbation

$\Phi_P = \delta P_g$.

It does not follow automatically from the static weak-field equation

$\nabla^2 \Phi_g = (4\pi G/c^2)(E_d + 3P_g)$.

Therefore, the precise theoretical statement is:

The PPC framework is extended by postulating that gravitational-pressure perturbations constitute a massless scalar field obeying the covariant wave equation $\Box_g \Phi_P = 0$. In the weak-field limit, this equation reduces to $\partial^2 \Phi_P / \partial t^2 = c^2 \nabla^2 \Phi_P$.

This distinction should be maintained when presenting the scalar-wave formulation as a research hypothesis.

25. CONCLUSION

This paper develops a separate scalar gravitational-pressure wave formulation within the PPC Law of Gravity.

The PPC energy density is

$$E_d = \rho c^2$$

and gravitational pressure is

$$P_g = \omega E_d.$$

The PPC weak-field gravitational equation is

$$\nabla^2 \Phi_g = (4\pi G/c^2)(E_d + 3P_g).$$

A perturbation in gravitational pressure is defined by

$$\Phi_P = \delta P_g.$$

The proposed covariant scalar-wave equation is

$$\Box_g \Phi_P = 0.$$

In the weak-field limit,

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$$

with

$$|h_{\mu\nu}| << 1,$$

the scalar equation becomes

$$\Box \Phi_P = 0.$$

Therefore, the explicit scalar gravitational-pressure wave equation is

$$\partial^2 \Phi_P / \partial t^2 = c^2 \nabla^2 \Phi_P.$$

The formulation distinguishes three important quantities:

Φ_g = static gravitational potential

$h_{\mu\nu}$ = tensorial metric perturbation

Φ_P = scalar gravitational-pressure perturbation.

The PPC gravitational pressure contributes to the curvature source, while its gradient contributes to pressure-force dynamics.

The proposed scalar sector therefore provides a possible dynamical extension of PPC gravity in which gravitational-pressure perturbations propagate as scalar waves in the weak-field limit.

FINAL CENTRAL EQUATIONS

$E_d = \rho c^2$

$P_g = \omega E_d$

$\nabla^2 \Phi_g = (4\pi G/c^2)(E_d + 3P_g)$

$P_g = P_{g0} + \Phi_P$

$\Phi_P = \delta P_g$

$\Box_g \Phi_P = 0$

$\Box \Phi_P = 0$

$\partial^2 \Phi_P / \partial t^2 = c^2 \nabla^2 \Phi_P$

$\nabla P_g = \nabla \Phi_P$

$\Phi_P = A \exp[i(k \cdot r - \omega t)]$

$\omega^2 = c^2 k^2$

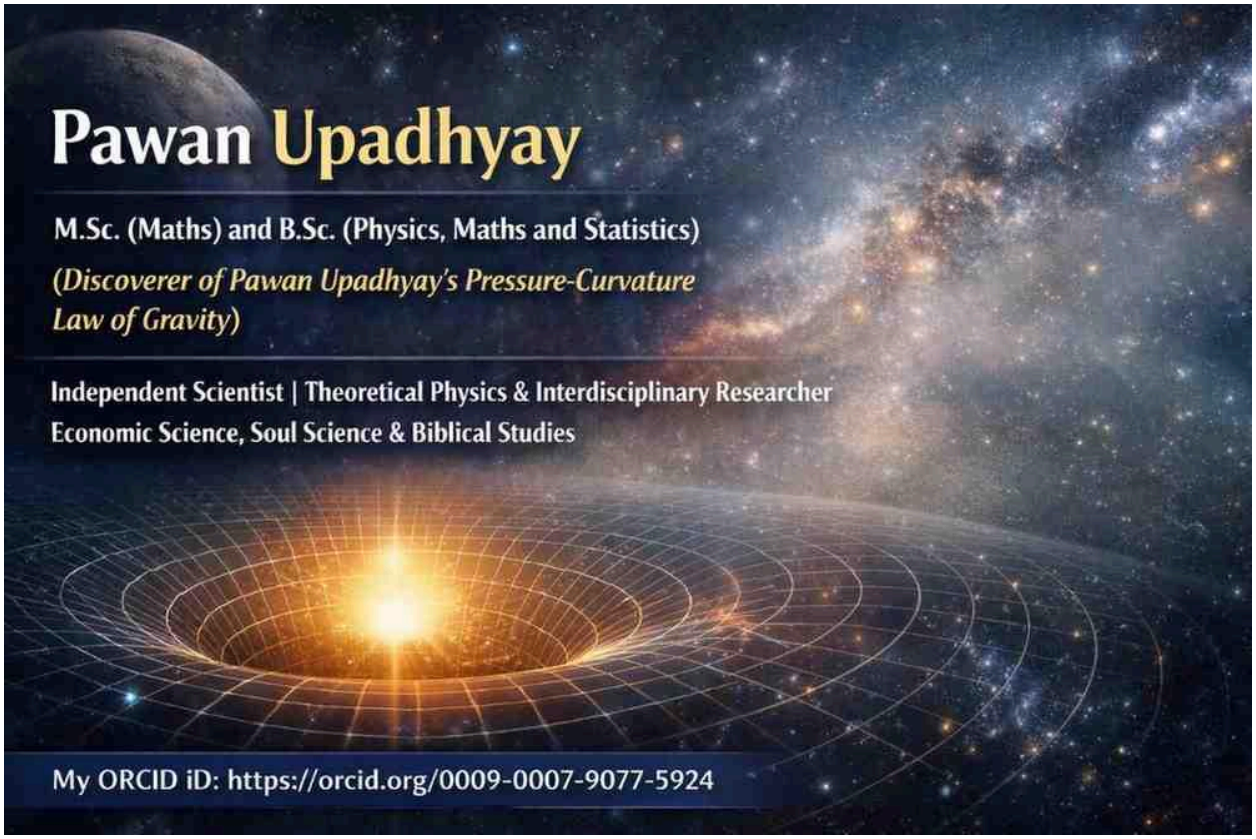
$v = c.$

3rd Research Paper

LINEARIZED EINSTEIN EQUATION WITH THE PPC ENERGY-PRESSURE TENSOR

A Weak-Field Tensorial Formulation of the Pressure Curvature Law of Gravity

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ABSTRACT

The weak-field limit of general relativity provides a linearized description of spacetime perturbations around Minkowski spacetime. In this formulation, the metric is written as a background Minkowski metric plus a small perturbation, and the Einstein field equations reduce to a linear wave equation for the trace-reversed metric perturbation.

The Pressure Curvature Law (PPC) of Gravity introduces gravitational pressure P_g together with energy density E_d as fundamental gravitational source quantities. The PPC energy-pressure tensor is

$$T_{\mu\nu}^{PPC} = (E_d + P_g)u_\mu u_\nu + P_g g_{\mu\nu},$$

with

$$E_d = \rho c^2$$

and

$$P_g = \omega E_d.$$

This paper investigates the direct substitution of the PPC tensor into the linearized Einstein equation. The resulting equation is

$$\Box \bar{h}_{\mu\nu} = -(16\pi G/c^4)[(E_d + P_g)u_\mu u_\nu + P_g g_{\mu\nu}].$$

Using the PPC relation $P_g = \omega E_d$ gives

$$\Box \bar{h}_{\mu\nu} = -(16\pi G/c^4)[(1 + \omega)E_d u_\mu u_\nu + \omega E_d g_{\mu\nu}].$$

This formulation provides a tensorial weak-field representation in which both energy density and gravitational pressure contribute directly to the source of metric perturbations. In the vacuum limit, $E_d \to 0$ and $P_g \to 0$, the equation reduces to the standard homogeneous linearized gravitational-wave equation

$$\Box \bar{h}_{\mu\nu} = 0.$$

The paper establishes the mathematical structure of this PPC-modified source formulation and distinguishes it from the separate scalar gravitational-pressure wave equation.

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1. INTRODUCTION

General relativity describes gravitation through the Einstein field equation

$$G_{\mu\nu} = (8\pi G/c^4)T_{\mu\nu}.$$

The full Einstein equation is nonlinear. Its weak-field approximation is obtained by considering a spacetime metric close to Minkowski spacetime:

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$$

with

$|h_{\mu\nu}| \ll 1$.

In the linearized theory, the gravitational field is represented by the tensorial metric perturbation $h_{\mu\nu}$.

The PPC Law of Gravity introduces a modified energy-pressure description in which gravitational pressure P_g contributes together with energy density E_d.

The PPC energy-pressure tensor is

$T_{\mu\nu}^{PPC} = (E_d + P_g)u_\mu u_\nu + P_g g_{\mu\nu}$.

The PPC framework defines

$E_d = \rho c^2$

and

$P_g = \omega E_d$.

The objective of this paper is to investigate what results when this PPC tensor is used as the source in the linearized Einstein equation.

---

2. FULL EINSTEIN FIELD EQUATION

The Einstein field equation is

$G_{\mu\nu} = (8\pi G/c^4)T_{\mu\nu}$.

In this equation:

$G_{\mu\nu}$ = Einstein tensor

$T_{\mu\nu}$ = energy-momentum tensor

G = gravitational constant

c = speed of light.

The full equation is nonlinear because $G_{\mu\nu}$ depends nonlinearly on the metric and its derivatives.

The weak-field formulation is obtained by expanding the metric around Minkowski spacetime.

---

3. WEAK-FIELD METRIC

The weak-field metric is written as

$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$

where

$|h_{\mu\nu}| << 1$.

Here,

$\eta_{\mu\nu}$ = Minkowski metric

$h_{\mu\nu}$ = small metric perturbation.

The inverse metric is correspondingly expanded to first order:

$g^{\mu\nu} \approx \eta^{\mu\nu} - h^{\mu\nu}$.

Only terms linear in $h_{\mu\nu}$ are retained.

---

4. TRACE-REVERSED METRIC PERTURBATION

Define the trace of the metric perturbation as

$h = \eta^{\mu\nu}h_{\mu\nu}$.

The trace-reversed perturbation is

$\bar{h}_{\mu\nu} = h_{\mu\nu} - (1/2)\eta_{\mu\nu}h$.

The linearized Einstein equations can then be expressed in terms of $\bar{h}_{\mu\nu}$.

Under the Lorenz gauge condition,

$\partial^\mu \bar{h}_{\mu\nu} = 0$,

the linearized Einstein equation takes the form

$\Box \bar{h}_{\mu\nu} = -(16\pi G/c^4) T_{\mu\nu}$.

This is the starting equation for introducing the PPC tensor.

---

5. PPC ENERGY-PRESSURE TENSOR

The PPC tensor is

$T_{\mu\nu}^{PPC} = (E_d + P_g) u_\mu u_\nu + P_g g_{\mu\nu}$.

The PPC energy density is

$E_d = \rho c^2$.

The gravitational pressure is

$P_g = \omega E_d$.

Therefore,

$T_{\mu\nu}^{PPC} = (E_d + \omega E_d) u_\mu u_\nu + \omega E_d g_{\mu\nu}$.

Hence,

$T_{\mu\nu}^{PPC} = (1 + \omega) E_d u_\mu u_\nu + \omega E_d g_{\mu\nu}$.

This expresses the PPC source entirely in terms of E_d, ω, u_μ, and $g_{\mu\nu}$.

---

6. PPC TENSOR IN THE LINEARIZED EINSTEIN EQUATION

The linearized Einstein equation is

$\Box \bar{h}_{\mu\nu} = -(16\pi G/c^4) T_{\mu\nu}$.

Replace the conventional source tensor with the PPC tensor:

$T_{\mu\nu} \rightarrow T_{\mu\nu}^{PPC}$.

Therefore,

$\Box \bar{h}_{\mu\nu} = -(16\pi G/c^4) T_{\mu\nu}^{PPC}$.

Substituting the PPC tensor gives

$\Box \bar{h}_{\mu\nu} = -(16\pi G/c^4)[(E_d + P_g)u_\mu u_\nu + P_g g_{\mu\nu}]$.

This is the central equation of this paper.

---

7. PPC LINEARIZED EINSTEIN EQUATION USING $E_d = \rho c^2$

Using

$E_d = \rho c^2$,

the equation becomes

$\Box \bar{h}_{\mu\nu} = -(16\pi G/c^4)[(\rho c^2 + P_g)u_\mu u_\nu + P_g g_{\mu\nu}]$.

This equation shows explicitly that both the matter-energy density and gravitational pressure contribute to the source of the metric perturbation.

---

8. PPC LINEARIZED EQUATION USING $P_g = \omega E_d$

Using

$P_g = \omega E_d$,

we obtain

□h̄_μv = −(16πG/c⁴)[(1 + ω)E_d u_μu_v + ωE_d g_μv].

Factoring out E_d,

□h̄_μv = −(16πG E_d/c⁴)[(1 + ω)u_μu_v + ωg_μv].

Using

E_d = ρc²,

this becomes

□h̄_μv = −(16πGρ/c²)[(1 + ω)u_μu_v + ωg_μv].

This is the PPC-source form of the linearized Einstein equation.

---

9. PHYSICAL INTERPRETATION

The equation

□h̄_μv = −(16πG/c⁴)[(E_d + P_g)u_μu_v + P_g g_μv]

has two sides.

The left-hand side,

□h̄_μv,

describes the propagation and spatial-temporal behavior of the tensorial metric perturbation.

The right-hand side,

−(16πG/c⁴)T_μv^PPC,

describes the PPC source generating the perturbation.

Thus,

PPC energy-pressure source

↓

T_μv^PPC

↓

metric perturbation

↓

ħ_μv.

The gravitational pressure P_g is therefore included directly in the source of the tensorial weak-field gravitational field.

---

10. ROLE OF GRAVITATIONAL PRESSURE

The PPC framework gives gravitational pressure two important roles.

First,

P_g → curvature source.

Second,

∇P_g → pressure-force dynamics.

The PPC material explicitly distinguishes these two roles.

In the linearized tensor equation, the first role appears directly:

P_g → T_μv^PPC → □ħ_μv.

The second role remains associated with the pressure gradient:

∇P_g.

Therefore, the tensorial weak-field equation does not eliminate the dynamical pressure-gradient mechanism.

---

11. LINEARIZED PPC FIELD EQUATION

The proposed tensorial PPC weak-field equation is

$\Box \bar{h}_{\mu\nu} = -(16\pi G/c^4)[(E_d + P_g)u_\mu u_\nu + P_g g_{\mu\nu}]$.

Using

$P_g = \omega E_d$,

it becomes

$\Box \bar{h}_{\mu\nu} = -(16\pi G\, E_d/c^4)[(1 + \omega)u_\mu u_\nu + \omega g_{\mu\nu}]$.

Using

$E_d = \rho c^2$,

we obtain

$\Box \bar{h}_{\mu\nu} = -(16\pi G\rho/c^2)[(1 + \omega)u_\mu u_\nu + \omega g_{\mu\nu}]$.

These three equations are mathematically equivalent under the PPC definitions.

---

12. VACUUM LIMIT

Consider the limit

$E_d \to 0$.

Since

$P_g = \omega E_d$,

we also have

$P_g \to 0$.

Therefore,

$T_{\mu\nu}^{PPC} \to 0$.

The linearized PPC equation becomes

$\Box \bar{h}_{\mu\nu} = 0$.

Thus, the vacuum limit of the PPC-source formulation is the homogeneous tensorial wave equation.

The sequence is

$E_d \to 0$

$\downarrow$

$P_g \to 0$

$\downarrow$

$T_{\mu\nu}^{PPC} \to 0$

$\downarrow$

$\Box \bar{h}_{\mu\nu} = 0.$

This establishes the mathematical vacuum limit of the proposed source substitution.

---

13. TENSORIAL GRAVITATIONAL WAVES

In vacuum, the equation

$\Box \bar{h}_{\mu\nu} = 0$

admits wave solutions.

A plane-wave solution can be written schematically as

$\bar{h}_{\mu\nu} = A_{\mu\nu} \exp[i(k \cdot x)].$

The wave vector satisfies the null condition

$k_\mu k^\mu = 0.$

Therefore, the tensorial perturbation propagates at

$c.$

The important point is that $\bar{h}_{\mu\nu}$ remains a tensor field.

The PPC modification in this paper occurs through its source:

$T_{\mu\nu} \to T_{\mu\nu}^{PPC}.$

---

14. PPC PRESSURE PERTURBATION

Now consider a perturbation in gravitational pressure:

$P_g = P_{g0} + \delta P_g$.

Define

$\Phi_P = \delta P_g$.

Then

$P_g = P_{g0} + \Phi_P$.

The perturbation of the PPC tensor can be written, under a direct pressure-only perturbation with the other quantities held fixed, as

$\delta T_{\mu\nu}^{PPC} = \Phi_P(u_\mu u_\nu + g_{\mu\nu})$.

The corresponding perturbation of the linearized source is

$\delta(\Box \bar{h}_{\mu\nu}) = -(16\pi G/c^4)\delta T_{\mu\nu}^{PPC}$.

Thus,

$\delta(\Box \bar{h}_{\mu\nu}) = -(16\pi G/c^4)\Phi_P(u_\mu u_\nu + g_{\mu\nu})$.

This provides a possible coupling between a scalar pressure perturbation and the tensorial metric perturbation.

---

15. TENSORIAL-SCALAR STRUCTURE

The tensorial sector is

$\Box \bar{h}_{\mu\nu} = -(16\pi G/c^4)T_{\mu\nu}^{PPC}$.

The scalar pressure perturbation is

$\Phi_P = \delta P_g$.

If a scalar wave equation is additionally postulated,

□_gΦ_P = 0.

In the weak-field limit,

□Φ_P = 0.

Therefore,

∂²Φ_P/∂t² = c²∇²Φ_P.

The resulting structure is

Φ_P → δP_g → δT_μν^PPC → δh̄_μν.

Thus, the scalar pressure field can act as a source perturbation for the tensorial gravitational field.

---

16. DIFFERENCE BETWEEN THE TWO WAVE EQUATIONS

The tensorial equation is

□h̄_μν = 0

in vacuum.

The scalar equation is

□Φ_P = 0.

Although both have the same mathematical wave operator, they represent different fields.

h̄_μν is tensorial.

Φ_P is scalar.

Therefore,

□h̄_μν ≠ □Φ_P

in terms of the physical field being propagated, even though both equations use the d'Alembertian operator.

The proposed PPC scalar wave should therefore be treated as a separate scalar degree of freedom rather than being identified with the tensorial metric perturbation.

---

17. RELATION TO THE PPC STATIC WEAK-FIELD EQUATION

The PPC static weak-field equation is

$$\nabla^2 \Phi_g = (4\pi G/c^2)(E_d + 3P_g).$$

Using

$$P_g = \omega E_d,$$

we obtain

$$\nabla^2 \Phi_g = (4\pi G/c^2)E_d(1 + 3\omega).$$

Using

$$E_d = \rho c^2,$$

we obtain

$$\nabla^2 \Phi_g = 4\pi G\rho(1 + 3\omega).$$

This static equation and the tensorial linearized equation describe related but distinct weak-field descriptions.

The static equation describes the gravitational potential.

The linearized Einstein equation describes the tensorial metric perturbation.

The scalar pressure-wave equation describes a proposed dynamical perturbation of P_g.

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18. THREE LEVELS OF THE PPC WEAK-FIELD DESCRIPTION

The PPC weak-field framework can therefore be organized into three levels.

LEVEL 1 — STATIC SCALAR POTENTIAL

$\nabla^2\Phi_g = (4\pi G/c^2)(E_d + 3P_g)$

LEVEL 2 — TENSORIAL METRIC PERTURBATION

$\Box\bar{h}_{\mu\nu} = -(16\pi G/c^4)[(E_d + P_g)u_\mu u_\nu + P_g g_{\mu\nu}]$

LEVEL 3 — SCALAR PRESSURE PERTURBATION

$\Box\Phi_P = 0$

or

$\partial^2\Phi_P/\partial t^2 = c^2\nabla^2\Phi_P.$

These equations should not be treated as identical. They describe three different mathematical aspects of the proposed weak-field PPC formulation.

---

19. COMPLETE PPC LINEARIZED STRUCTURE

The proposed tensorial structure can be summarized as

$E_d = \rho c^2$

↓

$P_g = \omega E_d$

↓

$T_{\mu\nu}^{PPC} = (E_d + P_g)u_\mu u_\nu + P_g g_{\mu\nu}$

↓

$\Box \bar{h}_{\mu\nu} = -(16\pi G/c^4) T_{\mu\nu}^{PPC}$

↓

$\Box \bar{h}_{\mu\nu} = -(16\pi G/c^4)[(E_d + P_g)u_\mu u_\nu + P_g g_{\mu\nu}]$.

If the pressure perturbation is introduced,

$P_g = P_{g0} + \Phi_P$,

then

$\Phi_P = \delta P_g$

and

$\delta T_{\mu\nu}^{PPC} = \Phi_P(u_\mu u_\nu + g_{\mu\nu})$.

The scalar wave equation can then be postulated as

$\Box_g \Phi_P = 0$.

In the weak-field limit,

$\Box \Phi_P = 0$.

Therefore,

$\partial^2 \Phi_P / \partial t^2 = c^2 \nabla^2 \Phi_P$.

---

20. DISCUSSION

The important mathematical result of this paper is that the PPC tensor can be inserted directly into the source term of the linearized Einstein equation.

Starting from

$\Box \bar{h}_{\mu\nu} = -(16\pi G/c^4) T_{\mu\nu}$,

and using

$T_{\mu\nu} = T_{\mu\nu}^{PPC}$,

gives

□h̄_μv = −(16πG/c⁴)[(E_d + P_g)u_μu_v + P_g g_μv].

This does not change the tensorial nature of the linearized Einstein equation. Instead, it changes the assumed source tensor.

The PPC pressure therefore appears directly in the tensorial source.

Using

P_g = ωE_d

shows that the relative contribution of pressure is controlled by ω.

The equation becomes

□h̄_μv = −(16πG E_d/c⁴)[(1 + ω)u_μu_v + ωg_μv].

This provides a compact mathematical expression for the proposed PPC source in the weak-field tensorial framework.

---

21. THEORETICAL STATUS

It is important to distinguish the mathematical substitution from a derivation from standard general relativity.

The standard linearized Einstein equation has the structure

□h̄_μv = −(16πG/c⁴)T_μv.

The present paper proposes using the PPC tensor

T_μv^PPC = (E_d + P_g)u_μu_v + P_g g_μv

as the source.

Therefore, the resulting equation

□h̄_μv = −(16πG/c⁴)[(E_d + P_g)u_μu_v + P_g g_μv]

should be understood as the **PPC source formulation within the linearized Einstein framework**.

It is not claimed here that the PPC tensor follows automatically from the standard Einstein equations.

Likewise, the scalar equation

$\Box \Phi_P = 0$

is an additional proposed dynamical equation for gravitational-pressure perturbations.

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22. CONCLUSIONS

This paper has developed a separate weak-field tensorial formulation of the PPC Law of Gravity by inserting the PPC energy-pressure tensor into the linearized Einstein equation.

The PPC tensor is

$T_{\mu\nu}^{PPC} = (E_d + P_g)u_\mu u_\nu + P_g g_{\mu\nu}.$

With

$E_d = \rho c^2$

and

$P_g = \omega E_d,$

the PPC tensor becomes

$T_{\mu\nu}^{PPC} = (1 + \omega)E_d u_\mu u_\nu + \omega E_d g_{\mu\nu}.$

Substitution into the linearized Einstein equation produces

$\Box \bar{h}_{\mu\nu} = -(16\pi G/c^4)[(E_d + P_g)u_\mu u_\nu + P_g g_{\mu\nu}].$

Equivalently,

$\Box \bar{h}_{\mu\nu} = -(16\pi G\, E_d/c^4)[(1 + \omega)u_\mu u_\nu + \omega g_{\mu\nu}].$

Using

$E_d = \rho c^2,$

the equation becomes

$\Box \bar{h}_{\mu\nu} = -(16\pi G\rho/c^2)[(1 + \omega)u_\mu u_\nu + \omega g_{\mu\nu}].$

In the vacuum limit,

$E_d \to 0$,

$P_g \to 0$,

and therefore

$\Box \bar{h}_{\mu\nu} = 0$.

The resulting formulation establishes a clear tensorial weak-field framework in which PPC gravitational pressure contributes directly to the source of metric perturbations.

A separate scalar sector may be introduced through

$\Phi_P = \delta P_g$

with the proposed equation

$\Box_g \Phi_P = 0$,

which becomes

$\partial^2 \Phi_P / \partial t^2 = c^2 \nabla^2 \Phi_P$

in the weak-field limit.

Thus, the proposed PPC framework can be represented by a tensorial gravitational field sourced by the PPC energy-pressure tensor, together with a possible scalar gravitational-pressure perturbation.

FINAL EQUATIONS FOR COPY AND PASTE

PPC ENERGY DENSITY:

$E_d = \rho c^2$

PPC GRAVITATIONAL PRESSURE:

$P_g = \omega E_d$

PPC ENERGY-PRESSURE TENSOR:

$T_{\mu\nu}^{PPC} = (E_d + P_g) u_\mu u_\nu + P_g g_{\mu\nu}$

LINEARIZED EINSTEIN EQUATION:

$$\Box \bar{h}_{\mu\nu} = -(16\pi G/c^4)T_{\mu\nu}$$

PPC-SOURCED LINEARIZED EINSTEIN EQUATION:

$$\Box \bar{h}_{\mu\nu} = -(16\pi G/c^4)[(E_d + P_g)u_\mu u_\nu + P_g g_{\mu\nu}]$$

USING $P_g = \omega E_d$:

$$\Box \bar{h}_{\mu\nu} = -(16\pi G/c^4)[(1 + \omega)E_d u_\mu u_\nu + \omega E_d g_{\mu\nu}]$$

USING $E_d = \rho c^2$:

$$\Box \bar{h}_{\mu\nu} = -(16\pi G\rho/c^2)[(1 + \omega)u_\mu u_\nu + \omega g_{\mu\nu}]$$

VACUUM LIMIT:

$$E_d \to 0$$

$$P_g \to 0$$

$$\Box \bar{h}_{\mu\nu} = 0$$

SCALAR PRESSURE PERTURBATION:

$$\Phi_P = \delta P_g$$

$$P_g = P_{g0} + \Phi_P$$

SCALAR WAVE EQUATION:

$$\Box_g \Phi_P = 0$$

WEAK-FIELD SCALAR WAVE EQUATION:

$$\Box \Phi_P = 0$$

EXPLICIT FORM:

$$\partial^2 \Phi_P/\partial t^2 = c^2 \nabla^2 \Phi_P$$

STATIC PPC WEAK-FIELD EQUATION:

$$\nabla^2 \Phi_g = (4\pi G/c^2)(E_d + 3P_g)$$

USING $P_g = \omega E_d$:

$\nabla^2\Phi_g = (4\pi G/c^2)E_d(1 + 3\omega)$

USING $E_d = \rho c^2$:

$\nabla^2\Phi_g = 4\pi G\rho(1 + 3\omega)$

Note: Gravitational Field and Gravitational-Pressure Field

1. Basic Concept

The **gravitational field** and the **gravitational-pressure field** are two different fields. They should not be identified as being the same field.

However, within the **Pawan Upadhyay's Pressure Curvature Law (PPC) framework**, they are mathematically connected through a set of physical and geometrical relations.

The distinction can therefore be expressed as:

Gravitational field ≠ Gravitational-pressure field

but

Gravitational field ↔ Gravitational-pressure field

through the PPC equations.

---

2. Gravitational-Pressure Field

The PPC framework introduces the gravitational pressure P_g in relation to energy density:

$E_d = \rho c^2$

and

$P_g = \omega E_d$

where:

- E_d = energy density
- ρ = mass density
- c = speed of light
- P_g = gravitational pressure
- ω = proportionality parameter

Thus, the gravitational-pressure field is associated with the distribution of energy density.

---

3. Pressure Gradient and Gravitational Interaction

The spatial variation of gravitational pressure produces a pressure gradient:

$$\nabla P_g$$

This provides the connection between the gravitational-pressure description and the gravitational interaction.

Schematically:

$$\rho \rightarrow E_d \rightarrow P_g \rightarrow \nabla P_g$$

In the relativistic PPC formulation, this pressure gradient is related to four-acceleration:

$$a^\mu = -[(g^\mu\nu + u^\mu u^\nu)\nabla_\nu P_g]/(E_d + P_g)$$

Therefore, the gravitational-pressure field and the gravitational field are not identical, but the **gradient of gravitational pressure provides a mathematical link to gravitational four acceleration**.

---

4. Weak-Field Connection

In the weak-field limit, the gravitational potential can be related to the PPC source terms through:

$$\nabla^2 \Phi_g = (4\pi G/c^2)(E_d + 3P_g)$$

where Φ_g represents the gravitational potential.

Using

$$E_d = \rho c^2$$

this becomes:

$$\nabla^2 \Phi_g = 4\pi G(\rho + 3P_g/c^2)$$

This equation demonstrates the mathematical connection between the gravitational potential and the gravitational-pressure field.

The important point is that Φ_g and P_g remain **different quantities**:

$$\Phi_g \neq P_g$$

but

$$\Phi_g \leftrightarrow P_g$$

through the field equation.

---

5. Scalar Gravitational-Pressure Perturbation

A perturbation of the gravitational-pressure field can be represented by a scalar field:

$$P_g = P_{g0} + \Phi_P$$

where

$$\Phi_P = \delta P_g$$

represents the perturbation of gravitational pressure.

If the background pressure is spatially uniform,

$$\nabla P_{g0} = 0$$

then

$$\nabla P_g = \nabla \Phi_P.$$

A proposed propagation equation for this scalar gravitational-pressure perturbation is:

$$\Box \Phi_P = 0$$

or, in flat spacetime,

$\partial^2\Phi_P/\partial t^2 = c^2\nabla^2\Phi_P$.

This scalar wave equation describes the **propagation of the gravitational-pressure perturbation**, rather than identifying the perturbation itself with the gravitational potential.

---

6. Tensorial and Scalar Descriptions

The PPC framework can therefore contain both tensorial and scalar mathematical descriptions.

Tensorial description

The metric perturbation is represented by $h_{\mu\nu}$, with the linearized Einstein equation in vacuum taking the form:

$\Box\bar{h}_{\mu\nu} = 0$

Here, $\bar{h}_{\mu\nu}$ is a tensorial gravitational-field perturbation.

Scalar description

The gravitational-pressure perturbation is represented by Φ_P:

$\Box\Phi_P = 0$

Thus:

$\bar{h}_{\mu\nu}$ → tensorial gravitational-field description

Φ_P → scalar gravitational-pressure perturbation

These should **not** be mathematically identified with each other.

---

7. Relationship Between the Two Fields

The relationship can be summarized as:

$E_d = \rho c^2$

$P_g = \omega E_d$

∇P_g

$\nabla^2 \Phi_g = (4\pi G/c^2)(E_d + 3P_g)$

and, for the proposed scalar pressure-wave sector,

$\Box \Phi_P = 0$.

In the relativistic regime:

$\nabla P_g \rightarrow a^\mu$

Therefore, the PPC framework provides a chain connecting the different descriptions:

Energy density → gravitational pressure → pressure gradient → gravitational potential / curvature → gravitational four acceleration

without requiring the gravitational field and gravitational-pressure field to be the same entity.

---

8. Key Statement

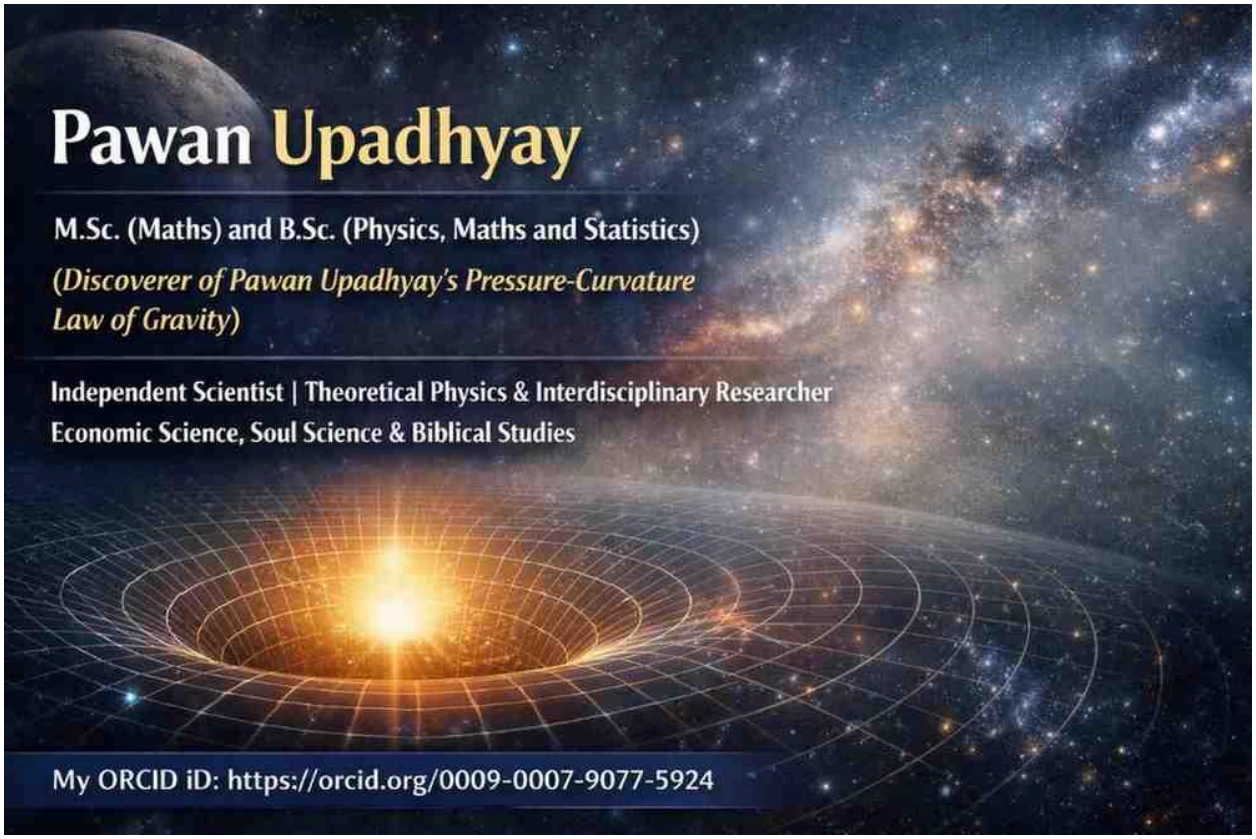
The gravitational field and gravitational-pressure field are distinct but interconnected fields. Their relationship is established through mathematical equations rather than by identifying one field with the other. In the weak-field regime, gravitational pressure contributes to the source of the gravitational potential, while its spatial gradient provides the connection to gravitational acceleration. In the relativistic regime, the pressure-gradient formulation can be expressed through four-acceleration.

4th Research Paper

INSERTION OF THE PPC METRIC TENSOR INTO THE LINEARIZED EINSTEIN EQUATION

A Weak-Field Tensorial Formulation of the Pressure Curvature Law of Gravity

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Abstract

This paper develops a specific mathematical formulation in which the **PPC metric tensor** is explicitly inserted into the **linearized Einstein equation**. The purpose is to establish a clear relationship between the PPC energy-pressure variables, the PPC metric, the metric perturbation, and the tensorial gravitational field equation.

The PPC framework defines the energy density as $E_d = \rho c^2$ and gravitational pressure through $P_g = \omega E_d$. Its energy-pressure tensor is written as $T_{\mu\nu} = (E_d + P_g)u_\mu u_\nu + P_g\, g_{\mu\nu}$, while the PPC metric is treated as a function of the PPC source variables, $g_{\mu\nu} = g_{\mu\nu}(E_d, P_g)$.

In the weak-field regime, the PPC metric is expanded around the Minkowski metric as $g_{\mu\nu}^{PPC} = \eta_{\mu\nu} + h_{\mu\nu}^{PPC}$, with $|h_{\mu\nu}^{PPC}| << 1$. Substitution into the PPC energy-pressure tensor produces a linearized source formulation in which the PPC metric explicitly participates in the stress-energy tensor. The resulting equation provides a tensorial weak-field representation of PPC gravity while maintaining a distinction between the gravitational-pressure field and the gravitational metric field.

---

1. Introduction

The Pressure Curvature Law of Gravity (PPC) introduces gravitational pressure P_g as a fundamental source variable associated with energy density.

The basic PPC relations are:

$$E_d = \rho c^2$$

and

$$P_g = \omega E_d.$$

The PPC formulation uses the energy-pressure tensor

$$T_{\mu\nu}^{PPC} = (E_d + P_g)u_\mu u_\nu + P_g g_{\mu\nu}.$$

This structure is explicitly given in the PPC formulation.

The PPC framework further specifies that the spacetime metric depends upon the PPC source variables:

$$g_{\mu\nu} = g_{\mu\nu}(E_d, P_g).$$

Thus, the proposed hierarchy is

$$(E_d, P_g) \rightarrow T_{\mu\nu} \rightarrow G_{\mu\nu} \rightarrow g_{\mu\nu}$$

and, through the metric connection,

$$(E_d, P_g) \rightarrow g_{\mu\nu} \rightarrow \Gamma^{\mu}_{\alpha\beta} \rightarrow \text{geodesic motion}.$$

The purpose of this paper is to examine what happens when the **PPC metric itself is explicitly inserted into the linearized Einstein equation**.

---

2. PPC Energy Density and Gravitational Pressure

The PPC energy density is defined by

$E_d = \rho c^2$

where ρ is the mass-density variable.

The gravitational pressure is related to energy density by

$P_g = \omega E_d$.

Therefore,

$P_g = \omega \rho c^2$.

These relations establish the PPC source variables:

$\rho \rightarrow E_d \rightarrow P_g$.

The PPC formulation treats P_g as a gravitational-pressure variable rather than merely introducing it as an unrelated conventional fluid pressure.

---

3. PPC Energy-Pressure Tensor

The PPC energy-pressure tensor is

$T_{\mu\nu}^{PPC} = (E_d + P_g)u_\mu u_\nu + P_g g_{\mu\nu}$.

The corresponding gravitational field equation is

$G_{\mu\nu} = (8\pi G/c^4)T_{\mu\nu}^{PPC}$.

Therefore,

$G_{\mu\nu} = (8\pi G/c^4)[(E_d + P_g)u_\mu u_\nu + P_g g_{\mu\nu}]$.

Using

$P_g = \omega E_d$

gives

$G_{\mu\nu} = (8\pi G/c^4)[(1 + \omega)E_d u_\mu u_\nu + \omega E_d g_{\mu\nu}]$.

Using

$E_d = \rho c^2$,

we obtain

$G_{\mu\nu} = (8\pi G\rho/c^2)[(1 + \omega)u_\mu u_\nu + \omega g_{\mu\nu}]$.

This is the tensorial PPC formulation used as the starting point for the weak-field treatment.

---

4. PPC Metric Tensor

The PPC framework specifies the metric as

$g_{\mu\nu} = g_{\mu\nu}(E_d, P_g)$.

This means that the metric is determined by the PPC gravitational source variables.

For a static spherical configuration, the PPC metric can be represented as

$ds^2 = -e^{2\nu(r)}c^2dt^2 + e^{2\lambda(r)}dr^2 + r^2(d\theta^2 + \sin^2\theta\, d\varphi^2)$.

Here,

$\nu = \nu(r)$

and

$\lambda = \lambda(r)$.

These functions are determined by the gravitational field equations and boundary conditions.

This metric therefore represents the geometrical sector of the PPC formulation.

---

5. Weak-Field Expansion of the PPC Metric

To insert the PPC metric into the **linearized Einstein equation**, the metric must first be expressed as a perturbation around flat spacetime.

We write

$$g_{\mu\nu}^{PPC} = \eta_{\mu\nu} + h_{\mu\nu}^{PPC}$$

where

$$\eta_{\mu\nu}$$

is the Minkowski metric and

$$h_{\mu\nu}^{PPC}$$

is the PPC metric perturbation.

The weak-field condition is

$$|h_{\mu\nu}^{PPC}| \ll 1.$$

Thus, the PPC metric is separated into two parts:

$$g_{\mu\nu}^{PPC} = \text{background metric} + \text{perturbation}$$

or

$$g_{\mu\nu}^{PPC} = \eta_{\mu\nu} + h_{\mu\nu}^{PPC}.$$

This distinction is essential because the **full PPC metric** and the **linearized metric perturbation** are not the same mathematical object.

---

6. Insertion of the PPC Metric into the Energy-Pressure Tensor

The PPC energy-pressure tensor is

$$T_{\mu\nu}^{PPC} = (E_d + P_g)u_\mu u_\nu + P_g\, g_{\mu\nu}^{PPC}.$$

Substituting

$$g_{\mu\nu}^{PPC} = \eta_{\mu\nu} + h_{\mu\nu}^{PPC}$$

gives

$$T_{\mu\nu}^{PPC} = (E_d + P_g)u_\mu u_\nu + P_g(\eta_{\mu\nu} + h_{\mu\nu}^{PPC}).$$

Expanding:

$$T_{\mu\nu}^{PPC} = (E_d + P_g)u_\mu u_\nu + P_g\eta_{\mu\nu} + P_g\, h_{\mu\nu}^{PPC}.$$

This is the central insertion developed in this paper.

It shows explicitly that the PPC metric contributes to the pressure portion of the energy-pressure tensor.

---

7. Linearized Einstein Equation

The linearized Einstein equation can be written as

$$\Box \bar{h}_{\mu\nu} = -(16\pi G/c^4)T_{\mu\nu}.$$

Substituting the PPC tensor gives

$$\Box \bar{h}_{\mu\nu}^{PPC} = -(16\pi G/c^4)[(E_d + P_g)u_\mu u_\nu + P_g\, g_{\mu\nu}^{PPC}].$$

Now insert

$$g_{\mu\nu}^{PPC} = \eta_{\mu\nu} + h_{\mu\nu}^{PPC}.$$

Therefore,

$$\Box \bar{h}_{\mu\nu}^{PPC} = -(16\pi G/c^4)[(E_d + P_g)u_\mu u_\nu + P_g\eta_{\mu\nu} + P_g\, h_{\mu\nu}^{PPC}].$$

This is the **PPC-metric-inserted linearized Einstein equation**.

---

8. Using the PPC Equation of State

Since

$P_g = \omega E_d$,

the equation becomes

$\Box \bar{h}_{\mu\nu}^{PPC} = -(16\pi G/c^4)[(1 + \omega)E_d u_\mu u_\nu + \omega E_d \eta_{\mu\nu} + \omega E_d h_{\mu\nu}^{PPC}]$.

Using

$E_d = \rho c^2$,

we obtain

$\Box \bar{h}_{\mu\nu}^{PPC} = -(16\pi G\rho/c^2)[(1 + \omega)u_\mu u_\nu + \omega\eta_{\mu\nu} + \omega h_{\mu\nu}^{PPC}]$.

This is a compact form of the proposed weak-field PPC tensor equation with the PPC metric explicitly inserted.

---

9. Order of the Metric-Perturbation Term

There is an important mathematical observation.

In the term

$P_g h_{\mu\nu}^{PPC}$

both P_g and $h_{\mu\nu}^{PPC}$ can be treated as perturbatively small quantities, depending on the weak-field expansion being adopted.

Consequently, their product may be a higher-order contribution.

If only **first-order perturbations** are retained, one may write

$P_g h_{\mu\nu}^{PPC} \approx 0$

at first order.

The equation then reduces to

$\Box \bar{h}_{\mu\nu}^{PPC} = -(16\pi G/c^4)[(E_d + P_g)u_\mu u_\nu + P_g\eta_{\mu\nu}]$.

Using the PPC relations:

$$\Box \bar{h}^{PPC}_{\mu\nu} = -(16\pi G\rho/c^2)[(1+\omega)u_\mu u_\nu + \omega\eta_{\mu\nu}].$$

Therefore, there are two useful levels of formulation.

Metric-retained form

$$\Box \bar{h}^{PPC}_{\mu\nu} = -(16\pi G/c^4)[(E_d + P_g)u_\mu u_\nu + P_g(\eta_{\mu\nu} + h^{PPC}_{\mu\nu})]$$

Strict first-order form

$$\Box \bar{h}^{PPC}_{\mu\nu} = -(16\pi G/c^4)[(E_d + P_g)u_\mu u_\nu + P_g\eta_{\mu\nu}]$$

The second is the appropriate form if the entire calculation is consistently truncated at first order.

---

10. Relationship Between the PPC Metric and Metric Perturbation

The relationship can be summarized as

$$g^{PPC}_{\mu\nu} = \eta_{\mu\nu} + h^{PPC}_{\mu\nu}.$$

The PPC metric is the full metric description.

The quantity $h^{PPC}_{\mu\nu}$ represents its weak-field perturbation.

Therefore:

$$g^{PPC}_{\mu\nu} \neq h^{PPC}_{\mu\nu}.$$

Instead,

$$h^{PPC}_{\mu\nu} = g^{PPC}_{\mu\nu} - \eta_{\mu\nu}.$$

This distinction is essential when constructing the linearized theory.

---

11. PPC Source-to-Geometry Relationship

The PPC framework proposes the following hierarchy:

$E_d \rightarrow P_g \rightarrow T_{\mu\nu}^{PPC} \rightarrow G_{\mu\nu} \rightarrow g_{\mu\nu}$.

The metric then determines the connection:

$g_{\mu\nu} \rightarrow \Gamma^{\mu}_{\alpha\beta}$.

The connection determines geodesic motion:

$\Gamma^{\mu}_{\alpha\beta} \rightarrow \text{geodesic motion}$.

Therefore:

$(E_d,P_g) \rightarrow T_{\mu\nu}^{PPC} \rightarrow G_{\mu\nu} \rightarrow g_{\mu\nu} \rightarrow \Gamma^{\mu}_{\alpha\beta} \rightarrow \text{motion}$.

This is consistent with the PPC formulation's stated field-to-geometry hierarchy.

---

12. Gravitational Field and Gravitational-Pressure Field

The PPC framework should distinguish two related but different concepts.

Gravitational-pressure field

The fundamental PPC pressure variable is

P_g.

Its spatial variation is

∇P_g.

Gravitational metric field

The geometrical gravitational field is represented by

$g_{\mu\nu}^{PPC}$

and, in the weak-field limit, by

$h_{\mu\nu}^{PPC}$.

Thus:

$P_g \neq g_{\mu\nu}$

and

$P_g \neq h_{\mu\nu}$.

However, the PPC framework connects them through the field equations:

$P_g \rightarrow T_{\mu\nu}^{PPC} \rightarrow G_{\mu\nu} \rightarrow g_{\mu\nu}$.

This distinction is important for the conceptual consistency of the model.

---

13. Weak-Field Gravitational Potential

The PPC formulation also uses the weak-field relation

$\nabla^2\Phi_g = (4\pi G/c^2)(E_d + 3P_g)$.

Using

$E_d = \rho c^2$,

this becomes

$\nabla^2\Phi_g = 4\pi G(\rho + 3P_g/c^2)$.

Thus the weak-field scalar potential description and the tensorial metric description can be regarded as different mathematical representations of the gravitational interaction.

The scalar potential Φ_g should not, however, be identified with the tensor perturbation $h_{\mu\nu}$.

---

14. Tensorial Description

The tensorial weak-field sector is represented by

$\bar{h}_{\mu\nu}^{PPC}$.

Its equation is

$\Box \bar{h}_{\mu\nu}^{PPC} = -(16\pi G/c^4) T_{\mu\nu}^{PPC}$.

The PPC source is

$T_{\mu\nu}^{PPC} = (E_d + P_g) u_\mu u_\nu + P_g\, g_{\mu\nu}^{PPC}$.

Thus the PPC pressure enters the tensorial gravitational equation through the energy-pressure tensor.

---

15. Scalar Pressure Description

A separate scalar perturbation may be introduced to represent changes in gravitational pressure:

$P_g = P_{g0} + \Phi_P$

where

$\Phi_P = \delta P_g$.

A proposed scalar propagation equation is

$\Box \Phi_P = 0$

or

$\partial^2 \Phi_P / \partial t^2 = c^2 \nabla^2 \Phi_P$.

This scalar equation should be treated as an **additional dynamical hypothesis of the PPC framework**, rather than as something automatically derived from the linearized Einstein equation.

Therefore:

$\bar{h}_{\mu\nu}^{PPC}$ → tensorial metric perturbation

Φ_P → scalar gravitational-pressure perturbation.

They are distinct mathematical quantities.

---

16. Static and Dynamic Sectors

The PPC formulation can consequently be separated into two related sectors.

Static weak-field sector

$\nabla^2\Phi_g = (4\pi G/c^2)(E_d + 3P_g)$.

Dynamic scalar pressure sector

$\Box\Phi_P = 0$.

Dynamic tensorial metric sector

$\Box\bar{h}_{\mu\nu}^{PPC} = -(16\pi G/c^4)T_{\mu\nu}^{PPC}$.

This gives three related but distinct descriptions:

Φ_g — gravitational potential

Φ_P — scalar pressure perturbation

$\bar{h}_{\mu\nu}^{PPC}$ — tensorial metric perturbation.

---

17. Vacuum Limit

For a vacuum exterior in which

$E_d = 0$

and

$P_g = 0$,

the PPC energy-pressure tensor becomes

$T_{\mu\nu}^{PPC} = 0$.

Consequently,

$\Box \bar{h}_{\mu\nu}^{PPC} = 0$.

This agrees with the vacuum form of the linearized Einstein equation.

The PPC formulation in the supplied research material likewise states that in vacuum $E_d = P_g = 0$, the PPC tensor equation reduces to

$G_{\mu\nu} = 0$.

This establishes the vacuum limit, although recovering a complete independent PPC prediction requires additional derivation beyond the vacuum equation itself.

---

18. Physical Interpretation

The proposed formulation gives the following interpretation:

E_d represents the energy-density source.

P_g represents gravitational pressure.

$T_{\mu\nu}^{PPC}$ represents the combined energy-pressure source.

$G_{\mu\nu}$ represents spacetime curvature.

$g_{\mu\nu}^{PPC}$ represents the PPC spacetime geometry.

$h_{\mu\nu}^{PPC}$ represents the weak-field metric perturbation.

∇P_g represents the spatial variation of gravitational pressure.

Therefore, the PPC source-to-geometry sequence can be written as:

$E_d \rightarrow P_g \rightarrow T_{\mu\nu}^{PPC} \rightarrow G_{\mu\nu} \rightarrow g_{\mu\nu}^{PPC}$.

The supplied PPC material similarly describes $(E_d + P_g)$ as controlling curvature magnitude and ∇P_g as determining curvature geometry.

---

19. Main Result

The principal equation developed in this paper is

$$\Box \bar{h}_{\mu\nu}^{PPC} = -(16\pi G/c^4)[(E_d + P_g)u_\mu u_\nu + P_g(\eta_{\mu\nu} + h_{\mu\nu}^{PPC})].$$

Since

$$g_{\mu\nu}^{PPC} = \eta_{\mu\nu} + h_{\mu\nu}^{PPC},$$

this can equivalently be written as

$$\Box \bar{h}_{\mu\nu}^{PPC} = -(16\pi G/c^4)[(E_d + P_g)u_\mu u_\nu + P_g\, g_{\mu\nu}^{PPC}].$$

Using

$$P_g = \omega E_d$$

and

$$E_d = \rho c^2,$$

the equation becomes

$$\Box \bar{h}_{\mu\nu}^{PPC} = -(16\pi G\rho/c^2)[(1 + \omega)u_\mu u_\nu + \omega g_{\mu\nu}^{PPC}].$$

For a strict first-order weak-field treatment, where the product $P_g\, h_{\mu\nu}^{PPC}$ is discarded as higher order:

$$\Box \bar{h}_{\mu\nu}^{PPC} = -(16\pi G\rho/c^2)[(1 + \omega)u_\mu u_\nu + \omega\eta_{\mu\nu}].$$

---

20. Important Theoretical Qualification

This formulation should be described carefully in a research paper.

The equation

$$\Box \bar{h}_{\mu\nu} = -(16\pi G/c^4)T_{\mu\nu}$$

is the **linearized Einstein equation**.

The substitution

$$T_{\mu\nu} \rightarrow T_{\mu\nu}^{PPC}$$

creates a **PPC source model within the linearized Einstein framework**.

It does not, by itself, demonstrate that PPC is an independent replacement for General Relativity.

Similarly, the assumption

$$g_{\mu\nu} = g_{\mu\nu}(E_d, P_g)$$

is a PPC model statement. To establish a complete independent PPC theory, one must derive the metric functions and their governing field equations from PPC itself and demonstrate that the resulting predictions differ from or reproduce GR where appropriate.

---

21. Conceptual Summary

The central structure of this paper is:

$$E_d = \rho c^2$$

$$P_g = \omega E_d$$

$$g_{\mu\nu}^{PPC} = g_{\mu\nu}(E_d, P_g)$$

$$g_{\mu\nu}^{PPC} = \eta_{\mu\nu} + h_{\mu\nu}^{PPC}$$

$$T_{\mu\nu}^{PPC} = (E_d + P_g)u_\mu u_\nu + P_g\, g_{\mu\nu}^{PPC}$$

$$\Box \bar{h}_{\mu\nu}^{PPC} = -(16\pi G/c^4) T_{\mu\nu}^{PPC}$$

Therefore:

$$(E_d, P_g) \rightarrow T_{\mu\nu}^{PPC} \rightarrow \bar{h}_{\mu\nu}^{PPC}$$

while simultaneously:

$$(E_d, P_g) \rightarrow g_{\mu\nu}^{PPC} \rightarrow \Gamma^{\mu}_{\alpha\beta} \rightarrow \text{geodesic motion}.$$

The gravitational-pressure field and gravitational metric field are therefore **different but mathematically connected**.

22. Conclusion

This paper has formulated an explicit insertion of the PPC metric tensor into the linearized Einstein equation.

The central construction is

$$g_{\mu\nu}^{PPC} = \eta_{\mu\nu} + h_{\mu\nu}^{PPC}$$

followed by

$$T_{\mu\nu}^{PPC} = (E_d + P_g)u_\mu u_\nu + P_g\, g_{\mu\nu}^{PPC}.$$

Substitution into the linearized Einstein equation gives

$$\Box \bar{h}_{\mu\nu}^{PPC} = -(16\pi G/c^4)[(E_d + P_g)u_\mu u_\nu + P_g\, g_{\mu\nu}^{PPC}].$$

The PPC relations

$$E_d = \rho c^2$$

and

$$P_g = \omega E_d$$

then give

$$\Box \bar{h}_{\mu\nu}^{PPC} = -(16\pi G\rho/c^2)[(1 + \omega)u_\mu u_\nu + \omega g_{\mu\nu}^{PPC}].$$

The strict first-order form is

$$\Box \bar{h}_{\mu\nu}^{PPC} = -(16\pi G\rho/c^2)[(1 + \omega)u_\mu u_\nu + \omega \eta_{\mu\nu}].$$

The resulting formulation preserves the distinction between **gravitational pressure** and **gravitational geometry**, while establishing a mathematical connection between them through the PPC energy-pressure tensor and the metric field.

The key conceptual chain is therefore:

Energy Density → Gravitational Pressure → PPC Energy-Pressure Tensor → Spacetime Curvature → PPC Metric → Metric Perturbation

or, more compactly,

$(E_d,P_g) \rightarrow T_{\mu\nu}^{PPC} \rightarrow G_{\mu\nu} \rightarrow g_{\mu\nu}^{PPC} \rightarrow h_{\mu\nu}^{PPC}$.

This provides a clear basis for treating the PPC metric as an explicit component of the weak-field tensorial formulation rather than treating the metric and gravitational-pressure fields as identical.

Note: Vacuum-Limit Recovery of the Einstein Gravitational-Wave Equation from the PPC Formulation

1. Purpose

This note examines whether the PPC energy-pressure tensor, when inserted into the linearized Einstein equation, can reproduce the **vacuum gravitational-wave equation** in the appropriate limit.

The result is affirmative at the level of the **vacuum field equation**.

---

2. PPC Energy-Pressure Tensor

The PPC energy-pressure tensor is

$$T_{\mu\nu}^{PPC} = (E_d + P_g)u_\mu u_\nu + P_g\, g_{\mu\nu}^{PPC}$$

where:

$$E_d = \rho c^2$$

and

$$P_g = \omega E_d.$$

The PPC formulation also describes the metric as dependent on the PPC source variables:

$$g_{\mu\nu}^{PPC} = g_{\mu\nu}(E_d,P_g).$$

---

3. Linearized Einstein Equation with the PPC Tensor

In the weak-field regime,

$$g_{\mu\nu}^{PPC} = \eta_{\mu\nu} + h_{\mu\nu}^{PPC}$$

where

$$|h_{\mu\nu}^{PPC}| \ll 1.$$

The linearized Einstein equation with the PPC source is

$$\Box \bar{h}_{\mu\nu}^{PPC} = -(16\pi G/c^4) T_{\mu\nu}^{PPC}.$$

Substituting the PPC tensor gives

$$\Box \bar{h}_{\mu\nu}^{PPC} = -(16\pi G/c^4)[(E_d + P_g)u_\mu u_\nu + P_g\, g_{\mu\nu}^{PPC}].$$

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4. Taking the Vacuum Limit

For the vacuum exterior, take

$$E_d = 0$$

and

$$P_g = 0.$$

Because

$$P_g = \omega E_d,$$

the second condition follows naturally when the PPC source energy density vanishes.

Therefore,

$$T_{\mu\nu}^{PPC} = (0 + 0)u_\mu u_\nu + 0 g_{\mu\nu}^{PPC}$$

and hence

$$T_{\mu\nu}^{PPC} = 0.$$

The linearized PPC equation consequently becomes

$$\Box \bar{h}^{PPC}_{\mu\nu} = 0.$$

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5. Comparison with the Einstein Vacuum Wave Equation

The vacuum linearized Einstein equation is

$$\Box \bar{h}_{\mu\nu} = 0.$$

Therefore, the PPC vacuum-limit equation has exactly the same mathematical form:

$$\Box \bar{h}^{PPC}_{\mu\nu} = 0$$

versus

$$\Box \bar{h}_{\mu\nu} = 0.$$

Thus:

$$\text{PPC vacuum limit} \rightarrow \text{Einstein linearized vacuum wave equation}.$$

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6. Interpretation

This result does **not** require the gravitational-pressure field to be identified with the gravitational metric field.

They remain distinct:

P_g → PPC gravitational-pressure variable

$g^{PPC}_{\mu\nu}$ → PPC spacetime metric

$h^{PPC}_{\mu\nu}$ → weak-field metric perturbation

The connection is through the PPC energy-pressure tensor:

$$(E_d, P_g) \rightarrow T^{PPC}_{\mu\nu} \rightarrow g^{PPC}_{\mu\nu} \rightarrow h^{PPC}_{\mu\nu}.$$

Your PPC formulation explicitly describes the source-to-geometry hierarchy as

$(E_d,P_g) \rightarrow T_{\mu\nu} \rightarrow G_{\mu\nu} \rightarrow g_{\mu\nu}$.

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7. What This Result Establishes

The result establishes:

In the vacuum limit $E_d = P_g = 0$, the PPC energy-pressure tensor vanishes, and the PPC-sourced linearized metric perturbation satisfies the homogeneous wave equation $\Box \bar{h}_{\mu\nu}^{PPC} = 0$, which has the same mathematical form as the vacuum linearized Einstein equation.

This is a legitimate **vacuum-limit recovery result**.

It also agrees with the PPC formulation in which the vacuum condition $E_d = P_g = 0$ reduces the PPC field equation to $G_{\mu\nu} = 0$.

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8. What It Does Not Yet Establish

The result should **not** be stated as proof that PPC independently derives the complete Einstein gravitational-wave theory.

The equation

$$\Box \bar{h}_{\mu\nu} = 0$$

is the vacuum propagation equation, but a complete comparison would additionally require analysis of the tensor perturbation, gauge conditions, polarization states, and observable wave solutions.

“The PPC formulation recovers the homogeneous linearized Einstein wave equation in the vacuum limit.”

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9. Proposed Research Claim

Claim: Vacuum Wave-Equation Recovery

$E_d = 0$

$$P_g = 0$$

$$\Rightarrow T_{\mu\nu}^{PPC} = 0$$

$$\Rightarrow \Box \bar{h}_{\mu\nu}^{PPC} = 0$$

Therefore,

$$\Box \bar{h}_{\mu\nu}^{PPC} = 0$$

recovers the mathematical form of the linearized Einstein vacuum wave equation.

This gives PPC framework a clear connection between its **energy-pressure source regime** and the **vacuum tensorial gravitational-wave regime**.